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Advances in High-Precision Interferometric Fiber Optic Gyroscope

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Abstract

As the core sensor of the inertial navigation system, the interferometric fiber optic gyroscope (IFOG) has gradually approached the theoretical limit in terms of precision performance after more than four decades of technological evolution, achieving a highest bias stability of $9.7\times10^{-7}{}^{\circ}/h$. Owing to its unique advantages, high-precision IFOGs have extended their applications beyond traditional navigation domains such as maritime, land, aerospace, and aviation, to encompass a variety of critical areas including precise Universal Time measurement, seismic monitoring, and micro-angle vibration detection. This work systematically reviews the key technological advancements in high-precision IFOGs, with a particular focus on improvements in precision performance, system component capabilities, and environmental adaptability. Moreover, it also explores future development trends toward higher precision, greater integration, and enhanced intelligence, aiming to provide valuable references for research and applications in related fields.

MAIN TEXT

1. Introduction

The fiber optic gyroscope, as one of the most representative applications of fiber-optic sensing technology, exemplifies the evolution of this field from theoretical research in laboratories to core components in advanced equipment [1–15]. Operating on the Sagnac-Laue effect, IFOGs achieve precise measurement of angular velocity by detecting the phase difference between counter-propagating light in a fiber coil. Since the first realization of an all-fiber interferometric structure in 1976 [16], IFOG technology has undergone several innovations. Evolving from the initial open-loop structure (with a precision of approximately 10°/h) to the digital closed-loop system (with a precision of less than 0.001°/h), its performance has improved by 7 to 8 orders of magnitude. It has become one of the irreplaceable core sensors in the fields of inertial navigation and precision guidance.

Currently, the complex space environment poses severe challenges to navigation autonomy. Factors such as rugged geographical environment, starlink interference, cyber-electronic warfare, and disruptions from low-Earth orbit satellite swarms have introduced risks of interference and failure for devices relying on the Global Positioning System (GPS). In this context, inertial navigation technology has emerged as the core solution for interference resistance due to its completely autonomous and continuous positioning ability. The performance bottleneck of system largely depends on the technological sophistication of the gyroscope. Depending on its unique performance advantages such as all-solid structure, wide precision coverage, large dynamic range, high reliability, long service life, long-term stability of parameters, fast start-up speed, high sensitivity, and small size, the IFOG promotes the dual breakthroughs in precision and reliability of the inertial system, becoming a key choice for navigation equipment. As the rapid iteration of inertial navigation technology and the expansion of extreme application environments, carrier platforms such as aircraft, ships, missiles, satellites, submarines, and unmanned aerial vehicles have put forward more stringent requirements for the performance of carrier attitude measurement in terms of precision, dynamic range, stability, and environmental adaptability. Therefore, the continuous research and application of high-performance IFOG are crucial for advancing the overall level of inertial navigation technology.

In response to the technological demand, the development directions of IFOG mainly focus on the precision enhancement and the stability optimization. The precision performance of IFOG is fundamentally determined by the performance indicators of core optical devices and the system's noise suppression capability, which provides a systematic approach for improving the comprehensive performance of high-precision IFOG. In addition to the optimization of device performance, the research on various precision enhancement and compensation techniques has also become a pivotal direction for technological breakthroughs in high-precision IFOG. Presently, the precision of IFOG that have been engineered and commercialized has covered the strategic grade. Under the leadership of companies such as Honeywell (US) and L3 Space & Navigation (US), iXblue (France), Optolink (Russia), and GEM elettronica (Italy), high-precision IFOGs have evolved from laboratory research into extensive practical applications. A relatively mature product system and market landscape have been established, and the distinctive products developed by these companies have collectively advanced the progress of the entire technological field. Notably, recent breakthroughs have elevated the precision of IFOG to the reference grade[17, 18], with the highest bias stability reaching $9.7\times10^{-7}{}^{\circ}/h$ [19], representing the highest level of current IFOG. This milestone provides strong support for applications requiring high-precision navigation, such as long-endurance autonomous navigation and high-resolution satellite earth observation systems, as well as for geophysical parameter measurement fields including precise Universal Time (UT1) measurement and seismic monitoring, and has important research value and broad application prospects.

Nowadays, the research on high-precision IFOG has entered a new stage of "quantum-classical" integration. On one hand, the theoretical phase accuracy can surpass the shot-noise limit and even the Heisenberg limit through quantum-enhanced interference using path-entangled NOON states [20]. On the other hand, relevant algorithms such as deep learning can be used to realize real-time compensation for various errors under complex operating conditions [21]. The sustained advancement of the technological path signifies the evolution of fiber-optic sensing technology from "precision measurement" to "extreme perception". This review focuses on introducing the research progress of high-precision IFOG in recent years, covering techniques for improving precision performance, enhancing the performance of system components, and increasing environmental adaptability, as shown in Fig. 1. Regarding the technology to improve precision performance, it mainly includes noise suppression technology, bias drift suppression technology, scale factor accuracy improvement technology, and some optimization technology. With respect to enhancing the performance of system components, the technology mainly targets the light source, multifunctional integrated phase modulator, and optical fiber. Moreover, the technology for increasing environmental adaptability describes techniques related to temperature and magnetic field adaptability. Finally, the 'Conclusion' section not only provides a comprehensive summary of the key points covered in this review but also puts forward significant potential directions for future research, which are expected to drive the further development of high-precision IFOG.

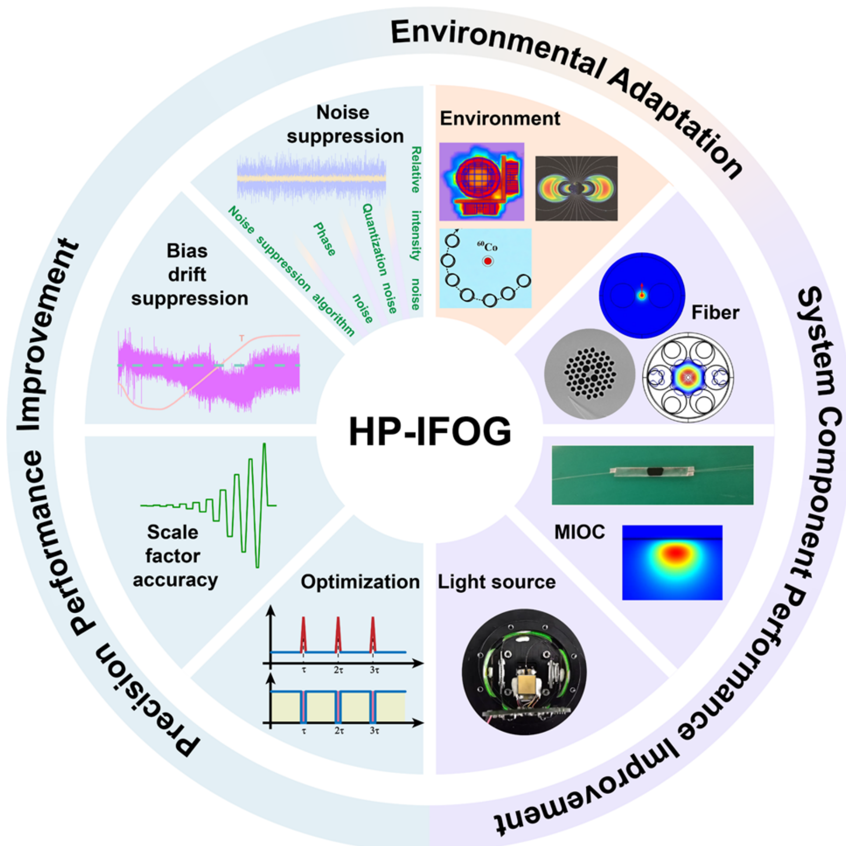

Fig. 1. Schematic diagram of the research progress on high-precision IFOGs' performance enhancement, mainly covering precision improvement techniques, system component performance enhancement techniques, and environmental adaptability improvement techniques. HP-IFOG: High-precision IFOG.

2. Precision Performance Improvement Technology

With the continuous increase in demands for measurement precision in modern inertial navigation systems, the precision of IFOG, as the core inertial sensor, has also been continuously improving. When high-precision IFOGs surpass the precision threshold of 0.001°/h, their signal characteristics exhibit significant differences from those of traditional navigation grade (0.01-0.1°/h) and tactical grade (0.1-10°/h) gyroscopes. The essence of this phenomenon is that the secondary error sources that could be ignored in the low-precision system begin to become the dominant factors restricting the precision improvement when the system noise floor is reduced to a specific level. They have a non-negligible impact on the signal-to-noise ratio (SNR) of high-precision IFOG. Therefore, the technology for improving the precision performance of IFOG mainly focuses on suppressing various error sources, so as to improve the SNR of signal measurement. Within the scope of our review, we survey the main relevant techniques involved in the process of achieving high-precision IFOGs and precision performance improvement.

2.1 Noise suppression technology

A. Relative intensity noise suppression technology

The suppression of relative intensity noise (RIN) in high-precision IFOGs is one of the key studies for improving the SNR and reducing the angular random walk (ARW) coefficient. RIN is determined by the spectral normalized autocorrelation function of the broadband light source and manifests as random fluctuations in optical power, directly affecting the ARW performance of IFOGs. As the optical power increases, the proportion of RIN caused by the light source in the total system noise gradually increases, becoming the main noise source affecting IFOGs [22]. Within a certain range, increasing the optical power can indeed improve the SNR. However, when the optical power exceeds a certain level, further increasing the light source power no longer significantly improves the SNR. Instead, it makes the impact of RIN on the back-end signal processing more prominent. Therefore, effectively suppressing RIN is essential for improving the precision performance of IFOGs.

As early as the 1990s, a method of suppressing RIN by subtraction in analog or digital electronic circuits was proposed [23,24], as illustrated in Fig. 2A. The approach is easy to implement and allows for precise control. However, it requires maintaining the consistency between the rotation signal and the reference signal in terms of polarization state, signal delay, and amplitude. To improve the robustness of the scheme, Mao Y et al. proposed a circuit noise subtraction method based on Wiener filter optimization [25]. The scheme uses a Wiener filter to reduce the degradation of the correlation of optical noise between the signal optical path and the reference optical path, which is caused by the imbalance of the detector and the signal processing circuit. As a result, the system noise suppression ratio reaches the limit of the optical system and the detection circuit. The method does not require additional hardware and can further improve the noise suppression level by optimizing the signal processing circuit. Thus, it provides a practical solution for the design of high-precision IFOGs. However, the approach relies heavily on pre-acquired data and is therefore unable to adapt to the dynamic variations of signal characteristics in time-varying environments, which may consequently lead to a degradation in system performance.

Moreover, F. Hakimi et al. proposed a scheme for suppressing the RIN of an incoherent broadband light source under the deep saturation state of a semiconductor optical amplifier (SOA) [26], as shown in Fig. 2B. The method leverages the nonlinear region caused by the gain saturation of the SOA to compress the optical power fluctuations, which is equivalent to a high-pass filter, thereby reducing the RIN. It is implied that the RIN suppression effect depends on the stability of the SOA. Relying on the method, the ARW coefficient in ultra-high-precision IFOGs can be reduced to 40% [27]. Ulteriorly, Liu X et al. explored three methods for suppressing the RIN in IFOGs [28], including the external modulation method using an intensity modulator (Fig. 2C), the SOA connection method, and a hybrid modulation scheme combining the two (Fig. 2D). They verified the suppression effects of these three methods through experiments. Among them, the external modulation method employs a negative feedback loop to feed the RIN back to a high-speed intensity modulator, thereby eliminating optical intensity fluctuations. The research found that the suppression effect of using the intensity modulator alone is better than that of using the SOA alone. Due to the inherent noise and stability issues of the intensity modulator and the SOA, the hybrid modulation scheme cannot achieve a complete superposition of the suppression effects. The suppression effectiveness of these three schemes depends on the performance of the added components. For instance, if the voltage noise introduced by the Direct-Current (DC) drift of the intensity modulator exceeds the RIN present in the detector, the system's negative feedback loop will fail to accurately compensate for the intensity fluctuations of the input light, thereby rendering the RIN suppression ineffective.

All-optical suppression technology is free from the limitations of the aforementioned schemes. In 2014, H. Lefèvre et al. from the iXBlue company proposed an optical technique for suppressing RIN [29]. By simply integrating a polarization-maintaining (PM) coupler and a PM isolator into the optical path of the IFOG to superpose the reference light and the signal light with the same intensity but a time delay, the suppression of RIN at the eigen frequency was achieved, as shown in Fig. 2E. To avoid the interference effect, a corresponding 90° PM fusion splicing was employed to ensure that the polarization states of the reference light and the signal light were orthogonal during superposition. The scheme significantly suppresses the RIN at the eigen frequency of the IFOG and has become the primary approach for suppressing RIN in high-precision IFOGs at present. After the implementation of all-optical technology, following the same suppression principle, the complex optical path structure described in [29] can be simplified. By merely incorporating a PM coupler, the suppression of RIN at the eigen frequency was also achieved [30], as shown in Fig. 2F. Compared with traditional IFOGs, only one optical element is added, which results in nearly a three-fold improvement in ARW. Meanwhile, the compact optical structure enhances the system's robustness, making it more adaptable to engineering applications. In 2018, Zheng Y et al. systematically studied the influence of different spectral errors (especially raised-cosine-type ripples) on the characteristics of RIN in IFOGs through theoretical analysis and experimental verification [31]. The study offers a theoretical basis for optimizing existing RIN suppression methods and contributes to improving the ARW of IFOGs by accurately evaluating the RIN power spectral density. In 2020, based on the research in [30], Zheng Y et al. continued to compress the optical structure of the RIN suppression system [32]. They inserted a Faraday rotator mirror into the dead arm of the source coupler (as shown in Fig. 2G) to realize the delay line filter for the RIN. They also discussed the main factors that affect the practical suppression effect, such as the unmatching of fiber lengths, the unbalancing of superposed intensities, and the inconsistency of the spectra. The scheme has a simple configuration and has potential reference value for the evaluation and optimization of other RIN suppression schemes. Futher, in 2023, Zheng Y et al. proposed a RIN suppression scheme based on self-adaptive adjustment of the modulation depth (Fig. 2H) to address the problem that the RIN suppression effect is ruined by the mismatching of the intensities at addition induced by external environments [33]. The robustness of the excess RIN suppression was significantly improved. However, the method has limitations such as insufficient flexibility in extracting the signal intensity and reference intensity, increased complexity of the control system, and the need for an additional design of a light intensity fault diagnosis algorithm. It is worth noting that all of the aforementioned all-optical schemes inherently double the optical intensity during photodetection, unavoidably introducing additional shot noise.

Then, in 2023, Chen Y et al. proposed a composite RIN suppression scheme combining the time-domain delay structure and the dual-polarization structure [34] and a cooperative four-state modulation scheme based on a source-sharing configuration [35]. The former scheme improves the ARW and bias stability of the IFOG by 8.65 and 8.06 times respectively, achieving a flat frequency response over a bandwidth of 0.05–100 Hz. However, although it combines the advantages of two traditional structures (the stability of the time-domain delay structure and the suppression effect of dual polarization structure), the system complexity increases significantly. The latter scheme achieves RIN compensation through reverse-phase square-wave modulation, significantly reducing the self-noise and ARW levels compared with the minimum configuration with conventional sinusoidal and square-wave modulation. In 2024, Chen Y et al. continued to study an improved PGC fusion algorithm combined with a source-sharing configuration scheme [36]. Through a simple and effective calculation method, the common-mode and differential-mode signals are effectively separated, achieving 16.5-fold self-noise suppression and 16-fold ARW suppression. Meanwhile, it has the ability to suppress environmental interference, enabling the IFOG to maintain high-precision performance in complex environments. While the three aforementioned schemes offer effective and comprehensive approaches to RIN handling, the dual-polarization design entails higher costs than conventional IFOGs.

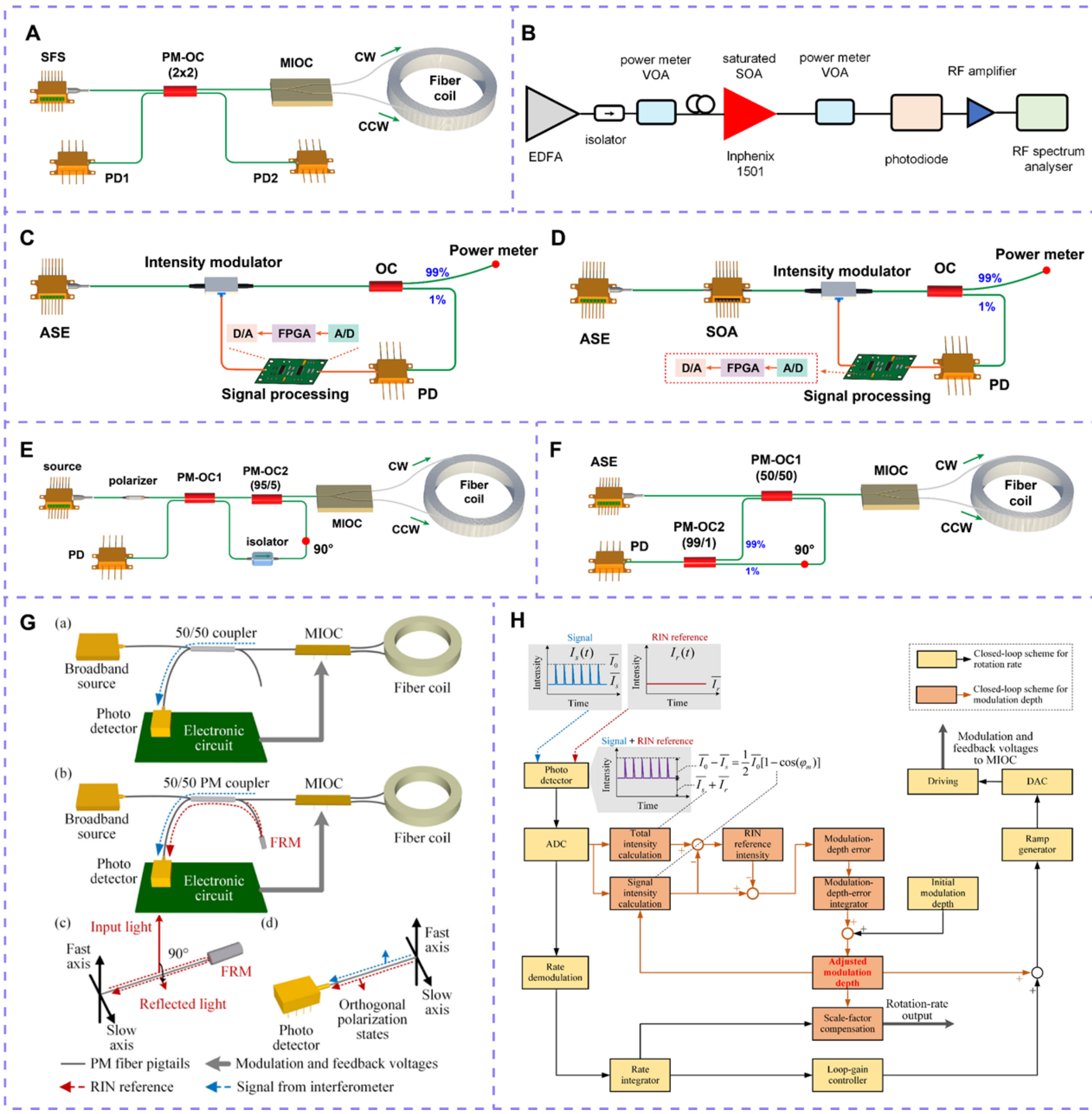

Fig. 2. Various schemes for RIN suppression technology. (A) Circuit subtraction scheme. SFS: superfluorescent fiber source. OC: Optical coupler. MIOC: Multifunction integrated optical chip. PD: Photodetector. CW: Clock Wise. CCW: Counter Clock Wise. (B) Gain saturation SOA scheme. Copyright 2020, The Institution of Engineering and Technology. (C) External modulation scheme for intensity modulator. ASE: Amplified spontaneous emission. (D) Hybrid modulation scheme of intensity modulator and SOA. (E) All-optical superposition scheme. (F) New all-optical superposition scheme. (G) All-optical

superposition scheme using an FRM (middle panel). Copyright 2020, IEEE. (H) Self-adaptive RIN suppression scheme. Copyright 2023, IEEE.

Feedback control may serve as a complementary approach for RIN suppression. In 2023, M. Kozuma et al. studied a new method to suppress RIN by using direct feedback to the superluminescent diode (SLD) drive current [37]. A portion of the light from the SLD is detected by a photodetector (PD), and the output voltage is used as a feedback signal to compensate for the RIN. The method can be extended to other light sources and can meet the strict requirements for ultra-high precision applications. Limited by the bandwidth of its feedback loop, the internal modulation scheme can only suppress the low-frequency components of RIN.

B. Quantization noise suppression technology

Quantization error is an inherent and unavoidable error in the digital closed-loop IFOG. It is mainly caused by the bit-length limitation limitation of electronic devices and data rounding in the digital signal processing. In medium and low precision IFOGs, the impact of quantization error is comparatively minor. However, as the precision of the gyroscope improves, the influence of this error on the detection precision rises markedly, especially in applications that require short-time and high-precision signal detection.

When an analog-to-digital converter (ADC) converts a continuous analog signal into a discrete digital one, the limited bit resolution allows the signal to be represented only as an integer multiple of the least significant bit (LSB), leading to rounding or truncation errors. Similarly, when a digital-to-analog converter (DAC) reconstructs the signal back into analog form, it also introduces quantization errors due to its finite bit-length. Moreover, during digital signal processing, limitations in operation bit-length and mismatches in data bit-length between modules can also generate additional quantization errors. To maintain the short-term detection precision of IFOGs, it is essential to implement effective strategies to suppress the impact of these quantization errors.

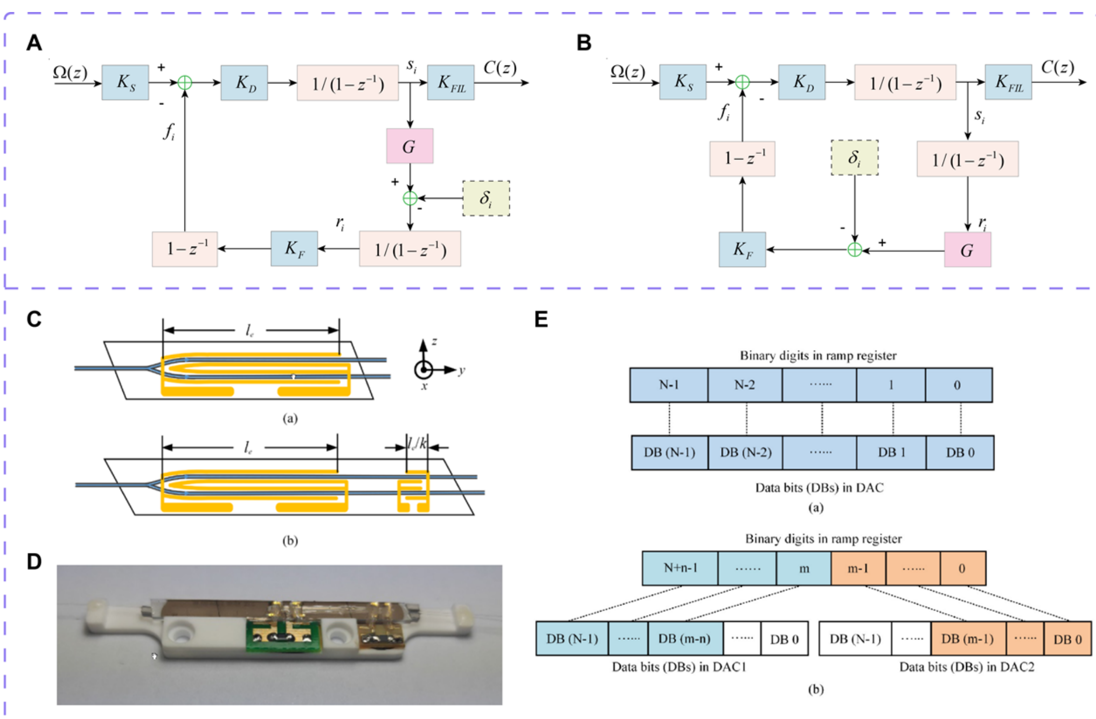

Fig. 3. Various schemes for quantization noise suppression technology. (A) The model of the FCLI scheme. The phase steps are first compressed and subsequently integrated to compensate for the modulation gain. (B) The model of the FILC scheme. The phase steps are first integrated and then compressed to compensate for the modulation gain. (C) The scheme of double-electrode-pair MIOC (bottom panel). The main electrode pair provides the modulation range, while the secondary pair determines the modulation precision, operating on the vernier principle to enhance the resolution of the feedback phase. (D) The product of double-electrode-pair MIOC. (E) Configuration of data bits in DACs for the double-electrode-pair MIOC (bottom panel). The (N+n)-bit digital ramp is split into the high-order bits and the low-order bits, which drive the main and secondary electrode pairs via two separate DACs, respectively. Reprinted with permission from [ref 40], copyright Optical Society of America.

An intuitive approach to reducing quantization error is to employ ADCs and DACs with higher bit resolution, thereby enhancing quantization precision. However, the method has notable limitations. On one hand, excessively high resolution will reduce the analog voltage corresponding to the LSB to the microvolt level, making it submerged by electronic noise and transient interference, which diminishes the effect of quantization error suppression. On the other hand, the dynamic performance of higher-resolution ADCs and DACs usually degrades, which in turn affects the precision performance of IFOGs. In 2017, to address the issue that the compensation algorithm of "First-Compression-Last-Integration (FCLI)" in the early real-time compensation research [38] of IFOG modulation errors was affected by quantization errors (Fig. 3A), Pan X et al. proposed a new digital compensation scheme, namely the "First-Integration-Last-Compression" (FILC) scheme [39]. By controlling the gain of the modulation chain and optimizing the sequence of integration/compression operations, they successfully achieved the suppression of quantization errors (Fig. 3B). In the scheme, the phase step between the modulation chain and the output chain is consistent, and the quantization error is suppressed through the differential operation of the optical path and time averaging, without any additional digital signal processing. Overall, this scheme offers a valuable reference for the suppression of quantization errors. In 2019, Zheng Y et al. proposed a double-electrode-pair multifunction integrated optical chip (MIOC) scheme for the suppression of quantization errors (Fig. 3C and D) [40]. By adding an additional pair of short electrodes, they significantly suppressed the quantization error, enabling the IFOG to achieve high-resolution measurements of extremely small angular rates in short smoothing times. The scheme opened up a new way to suppress quantization errors. Compared with the traditional MIOC structure with a single electrode pair, the design incorporating an additional pair of short electrodes can significantly increase the digital quantity of the feedback ramp (Fig. 3E). In the scheme, the quantization error is directly associated with the minimum quantization unit (1 LSB) of the feedback ramp. When the system maintains the same quantized analog voltage, the multi-electrode configuration enables a k-fold reduction in the generated quantization phase difference. This design enhances the quantization resolution of phase modulation, effectively improving the phase control precision of the system and achieving a smaller quantization error. However, the scheme requires the implementation of a more complex improved four-state modulation technique based on waveform decomposition and time-division. Furthermore, the electric fields between the dual electrode pairs may inadvertently introduce additional modulation, potentially generating new noise and degrading the overall system performance.

C. Phase noise suppression technology

As a phase-sensitive optical sensor based on the Sagnac effect, the IFOG's core measurement mechanism relies on the linear relationship between the phase difference of two counter-propagating light beams and the input angular velocity. However, this principle also makes the system highly susceptible to various types of phase errors. Any non-ideal phase disturbance can be directly converted into angular velocity measurement errors, leading to a degradation in system precision. For example, residual coherence phase errors and modulation phase errors can compromise long-term bias stability. Especially in high-precision application scenarios, random phase fluctuations such as thermal phase noise (TPN) can significantly degrade ARW performance. The coupling effect of these phase errors not only limits the long-term stability of IFOGs but also restricts their applicability in ultra-large-scale and dynamic environments. Consequently, the mechanism analysis and suppression technology of phase errors have become the critical research directions for improving the performance of IFOGs.

Residual coherence phase errors generally exist in dual-polarization IFOGs. In 2018, Peng C et al. investigated the impact of residual coherent phase errors on dual-polarization IFOGs [41], demonstrating that part of residual coherence phase errors cannot be compensated by weighted sum. To suppress these errors, they proposed two suppression methods: introducing sufficient temporal delay for the broadband light source or employing two independent but nearly identical light sources (Fig. 4A). Implementation of these two solutions yields IFOG bias instabilities of $5.53\times10^{-4}{}^{\circ}/h$ and $3.56\times10^{-4}{}^{\circ}/h$, respectively (Fig. 4B), with the latter demonstrating superior performance to the conventional single-polarization ‘minimal scheme’, albeit at the cost of requiring more optical components.

Modulation phase errors primarily originate from the modulation nonlinearity of the MIOC and are related to the driving voltage and temperature. In 2018, She X et al. proposed a novel method for measuring and compensating modulation phase errors [42]. Without altering the MIOC structure and manufacturing process, they achieved the suppression of modulation phase errors. The approach reduces the distortion of the detector signal by compensating the driving voltage (Fig. 4C) and uses the non-ideal driving voltage of the MIOC to correct the modulation phase, thereby achieving the flatness of the interference signal. In 2019, they further proposed a linear model of the MIOC modulation phase error based on the temperature characteristics of the equivalent capacitance [43]. For the first time, the model quantified the relationship between phase error and temperature (8.217 ppm/°C) (Fig. 4D), providing a theoretical foundation for temperature compensation in high-precision IFOGs.

Thermal phase noise, induced by random thermal fluctuations in the fiber, increases with the optical path length and has become a non-negligible noise source in high-precision IFOGs. In 2019, Peng C et al. revealed the spectral characteristics of TPN in a giant IFOG with a 30 km fiber coil through theoretical modeling and experimental validation [44]. Their study showed that TPN predominantly appears near the odd harmonic of the modulation frequency. They proposed to use high-order eigen frequency modulation technology to suppress the walk-off-terms of TPN, thereby reducing ARW, bias instability, and self-noise (Fig. 4E). In 2023, Ma X et al. proposed to use high-order frequency modulation technology to suppress TPN in large IFOGs [45]. By increasing the modulation frequency to a higher frequency band, they reduced the temperature sensitivity by 32 times without increasing the hardware costs, while maintaining compatibility with existing compensation methods. The approach significantly enhanced the temperature stability of the gyroscope, achieving a 3.8-fold reduction in ARW (Fig. 4F). However, due to the limitations imposed by the Nyquist sampling theorem, the modulation frequency cannot be increased indefinitely, and the suppression effect on ultra-high-frequency noise remains constrained. In the same year, the research group also studied a FOG fusion scheme based on a source-sharing architecture [46]. This scheme can simultaneously suppress the self-noise (including RIN and TPN) by 5.8 times and the bias instability by 5.2 times, providing a valuable reference for further research on the noise of IFOGs. However, the scheme entails a significantly higher cost due to its dual-gyro configuration.

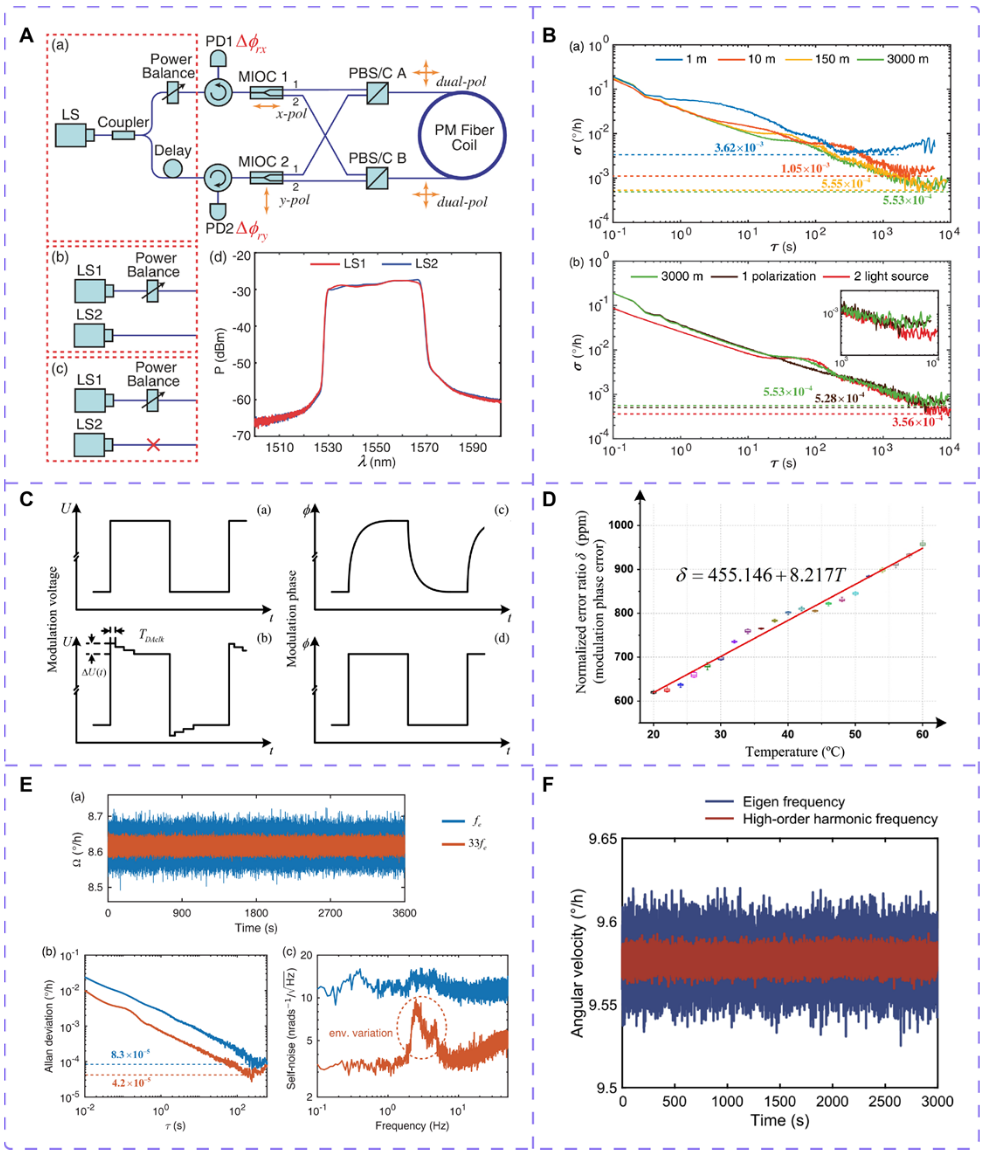

Fig. 4. Various schemes for phase noise suppression technology. (A) The scheme for dual-polarization IFOG to suppress residual coherent phase errors. Residual coherent phase error is suppressed by incorporating either an ultra-long fiber delay line or dual independent light sources, combined with a weighted summation approach. (B) Allan variance calculation results for residual coherent phase error suppression schemes. Copyright 2018 Optical Society of America. (C) The scheme for compensating eigen modulation voltage and eigen modulation phase (bottom panel). The driving waveform is modified to make the actual phase response closely approximate the ideal state. Copyright 2018, SPIE. (D) Quantitative relationship between modulation phase error and temperature. Copyright 2018, SPIE. (E) Test results of the giant IFOG performance improvement by high-order eigen frequency

modulation technology. Copyright 2019 Optical Society of America under the terms of the OSA Open Access Publishing Agreement. (F) Test result of the IFOG performance improvement by high-order frequency modulation technology. Rights managed by AIP Publishing.

D. Random noise suppression algorithm

The random errors of IFOGs are caused by uncalibrated interference factors such as environmental parameter fluctuations and electronic noise, which cannot be directly eliminated. Nevertheless, by establishing accurate error models and implementing effective compensation strategies, these random errors can be suppressed to within system tolerance limits. Consequently, random error suppression algorithms have become a pivotal component of noise suppression technology.

Digital filtering [47], wavelet transform [48], empirical mode decomposition (EMD) [49], and the Kalman filter (KF) [50,51] are widely used techniques for random noise suppression. However, each of these methods has inherent limitations. In recent years, to address the random drift errors in IFOGs, a hybrid algorithm combining the adaptive sampling strong tracking (ASST) algorithm with the scaled unscented Kalman filter (SUKF) has been proposed to solve the problem of signal denoising. By leveraging the innovation sequence and adaptive factors, it adaptively adjusts the state error covariance to achieve optimal performance, resulting in a 100-fold error suppression in the quiescent condition and improved mean-square error (MSE) performance in the manoeuvring case [52]. Despite its significant improvements in gyroscope precision, the method suffers from high algorithmic complexity and challenging hardware implementation. To overcome these drawbacks, the research group further developed the weighted covariance adaptive robust Kalman filter (WCARKF) algorithm [53], which introduces an adaptive tuning parameter and a weight function to adaptively measure the noise covariance matrix. By adjusting the weighted covariance of the innovation sequence and the adaptive parameter, the method achieves error suppression by several hundred times under the static case and improved root mean square error (RMSE) performance under the maneuvering case. Overall, it offers a more competitive solution compared to the approach in [52].

The improved double-factor adaptive Kalman filter (adaptive moving average–random weighting estimation–double-factor adaptive KF, AMA–RWE–DFAKF) algorithm enables effective noise reduction under both static and dynamic conditions, achieved through two independent adaptive mechanisms [54]. The first adaptive factor, the Kalman gain matrix, is updated in real time using random weighting estimation (RWE) based on the innovation covariance matrix. The second adaptive factor, the predicted state vector covariance matrix, is activated only when the adaptive moving average (AMA) detects discontinuities in the system state, and is also corrected using the RWE method. Through the double-factor adaptive adjustment, the algorithm achieves up to a 100-fold suppression of random error and demonstrates excellent root mean square error (RMSE) performance. Moreover, it offers a simpler implementation compared to the method in [52]. Similarly, the improved adaptive Kalman filter (KF) noise reduction algorithm based on innovation and RWE also adaptively updates the KF gain via the innovation covariance matrix estimated by RWE [55]. The difference is that to decrease the inertia of the KF response in the dynamic condition, an innovation-based chi-square test method is used to detect discontinuous IFOG signals, and the covariance matrix of the process noise is adjusted. The scheme provides superior noise suppression performance in both static and dynamic conditions, while also achieving lower computational complexity compared to the algorithm in [54]. However, it suffers from inherent limitations, particularly in terms of hardware implementation challenges.

The SSA-AND algorithm, which integrates singular spectrum analysis (SSA) with an augmented nonlinear differentiator (AND), is a novel denoising method for IFOG signals [56]. Compared with the widely used KF denoising algorithm, AND offers a simpler structure and operates without the need for pre-designed models. The SSA-AND algorithm decomposes the signal through singular spectrum analysis and combines the acceleration factor selection criterion of multi-scale transformation to obtain better denoising effects while ensuring no signal delay. The algorithm can effectively solve the contradiction between the convergence speed and noise suppression of AND and improve the reliability of the algorithm.

The denoising algorithm based on processing the intrinsic mode functions (IMF) obtained from EMD (EMD-IMF) can effectively address the problem that the traditional EMD method is prone to losing useful information [57]. By adaptively classifying IMF components through the fractal Gaussian noise energy model and the improved Hausdorff distance (HD) method and then processing different categories of IMF respectively, the algorithm achieves performance improvements such as an 87.7% reduction in bias drift and a 63.4% improvement in ARW. The scheme is fully data-driven and adaptive, offering strong generalizability. Moreover, integrating the approach with an improved EMD algorithm can further enhance noise reduction performance.

References [58] and [59] present the improved multi-scale permutation entropy complete ensemble empirical mode decomposition with adaptive noise (MPE-CEEMDAN) method and the adaptive filtering method based on quasi-proportional information (TVAKF) respectively. Both methods achieve superior random noise suppression and improve the precision performance of IFOGs in temperature-varying environments.

In addition, for the problem of analyzing the time-varying random errors of IFOGs, the dynamic Allan variance algorithm with an adaptive window length dynamically adjusts the window length by constructing an adaptive factor. The approach effectively resolves the limitation of traditional dynamic Allan variance, which struggles to balance dynamic tracking performance with the confidence of variance estimation [60]. The scheme can eliminate the influence of parameter deviations in complex environments and achieve high-precision online compensation for the random errors of IFOGs, providing an effective tool for the accurate evaluation of the above-mentioned various algorithms.

2.2 Bias drift suppression technology

Bias drift in IFOGs is a critical factor affecting their accuracy and stability, primarily resulting from optical effects, electronic and signal processing errors, and structural imperfections in the fiber. These factors induce the bias drift under changing environmental conditions such as temperature, magnetic fields, and radiation. This section focuses on mitigation schemes targeting the intrinsic causes of bias drift in IFOGs.

Electrical crosstalk error mainly originates from the electronic cross coupling between the signals of the feedback channel and the forward channel in IFOGs. Specifically, it occurs between the phase modulation voltage and the photodetector, and manifests as an additional bias error of IFOGs. If this error exceeds the input rate, it may lead to a dead zone in IFOGs. The most direct mitigation approach involves wiring the analog circuit and the digital circuit separately, strengthening the power supply filter, optimizing the layout of ground wire and taking the digital step wave along with the analog square wave to modulate the signal [61], thereby minimizing electronic crosstalk as much as possible. However, the solution requires substantial hardware modification and its effect varies depending on the precision level of the gyroscope. According to Reference [62], the additional bias error caused by electrical crosstalk depends on the correlation between the interference signal coupled into the photodetector and the demodulation sequence, which offers a new avenue for suppressing crosstalk-induced errors.

The random phase step technique is implemented by applying a random discrete voltage step to the feedback signal [63]. In the scheme, the voltage step is occasionally applied when forming the feedback ramp, with the period significantly shorter than the time before the feedback voltage could lock to a constant. (Fig. 5A). The approach avoids the electronic cross coupling caused by the fixed modulation waveform, and reduces the correlation between the interference signal and the demodulation sequence, thereby effectively suppressing electrical crosstalk errors. The random phase modulation technique [64] utilizes the random state-transition characteristic of the modulation signal and the randomness of the demodulated signal to ensure that the modulation sequence and the demodulation sequence are mutually independent. As a consequence, the correlation demodulation result between the modulation and demodulation sequences is orthogonal, thus achieving the suppression of electrical crosstalk errors (Fig. 5B). Statistically, these two random modulation schemes can completely eliminate electrical crosstalk errors. However, random modulation can lead to a reduction in the dynamic range of IFOGs and an increase in random noise, posing a potential risk of introducing low-frequency interference. In contrast, the six-state modulation (SSM) technique does not have the above-mentioned drawbacks. In 2018, Pan X et al. first proposed the six-state phase modulation scheme (Fig. 5C to E) [65]. By separating the fundamental frequencies of the modulation voltage and the demodulation reference signal, the correlation between them is reduced, thereby minimizing the bias error caused by electrical crosstalk. Theoretically, crosstalk can be completely eliminated by adjusting the modulation depth. Thus, the scheme is constrained by its reliance on RIN suppression techniques to attain a proper modulation depth. In 2024, a new six-state modulation scheme was proposed (Fig. 5F and G) [66]. The scheme uses a highly flexible modulation voltage to eliminate the correlation between the modulation voltage with a fixed phase delay and the demodulation sequence, effectively suppressing electrical crosstalk errors. Compared with the technique described in Reference [65], the scheme can eliminate the influence of crosstalk with a fixed phase delay at any modulation depth.

In addition, Shi H et al. proposed a periodic polarity conversion scheme to dynamically adjust the polarity of the Y-waveguide modulator [67], eliminating the bias error resulting from circuit cross-coupling. Meanwhile, based on the principle that a differential photodiode amplifier can effectively suppress coupling interference, Pan X et al. designed a high-performance fully differential photodiode amplifier (Fig. 5H) to tackle the issues of the large photodiode capacitance and parasitic capacitance, as well as the fact that the traditional three-op-amp instrumentation amplifier cannot meet the bandwidth requirements of IFOGs [68]. By applying reverse bias and using a modified tee-network to replace the feedback resistor, the amplifier suppresses electrical crosstalk errors to a great extent. These two schemes provide important technical references for suppressing electrical crosstalk errors in IFOGs.

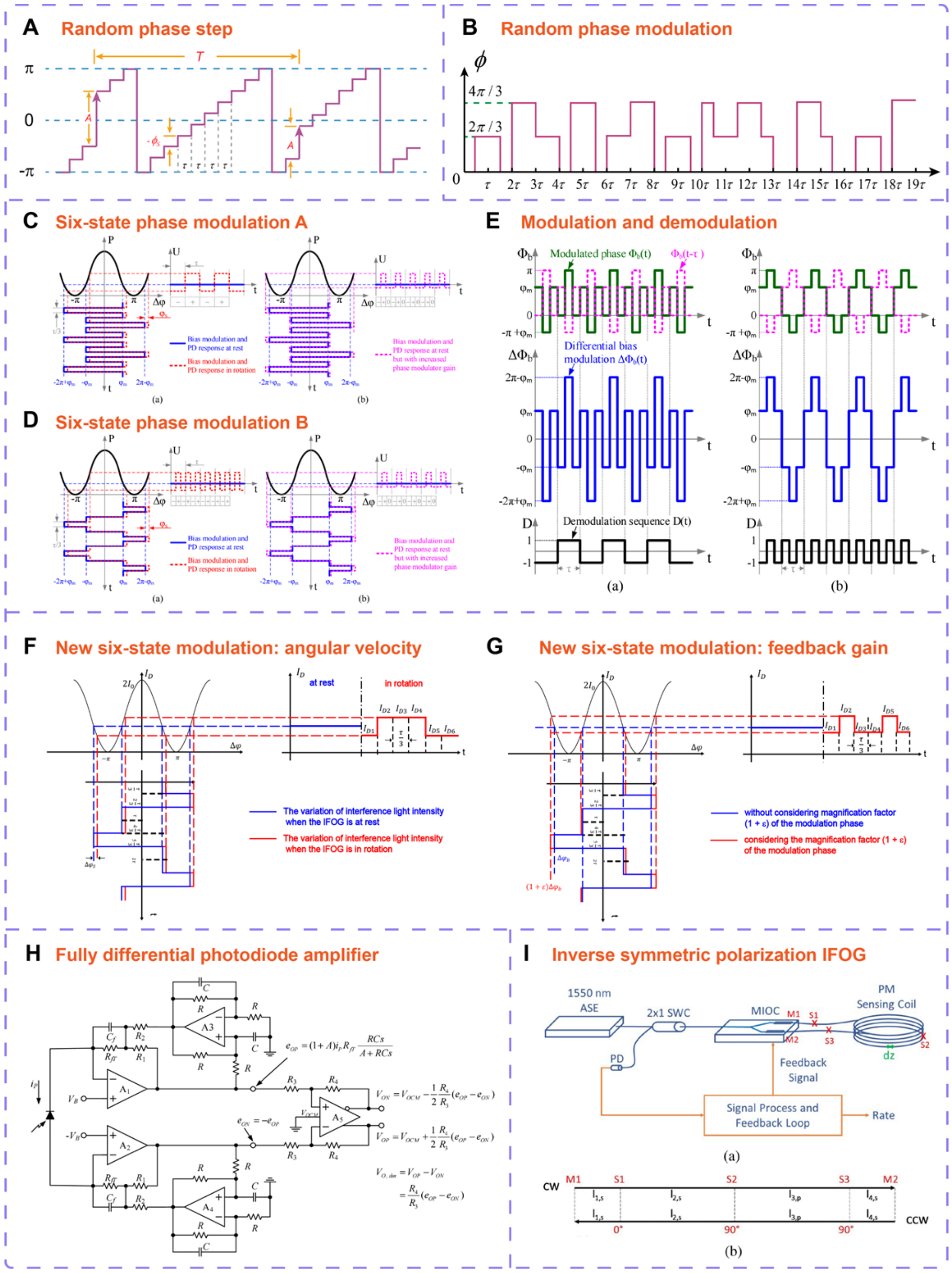

Fig. 5. Various schemes for bias drift suppression technology. (A) Ramp schematic of the random phase step technique. It shows the phase shift with the amplitude A and period T, in addition to the phase step. Random discrete voltage steps are applied to the feedback signal. τ : The transit time of light in the fiber coil. (B) Phase schematic of the random phase modulation technique. (C) Transfer function plot of an IFOG with six-state modulation A (SSMA). The sequence of six phase states within one modulation cycle is: $\varphi_m, -2\pi+\varphi_m, \varphi_m, -\varphi_m, 2\pi-\varphi_m, -\varphi_m$. (D) Transfer function plot of an IFOG with six-state modulation B (SSMB). The sequence of six phase states within one modulation cycle is: $\varphi_m, 2\pi-\varphi_m, \varphi_m, -\varphi_m, -2\pi+\varphi_m, -\varphi_m$. (E) Schematics of the modulation phase, modulation phase difference, and demodulation sequence for the SSMA and SSMB schemes. Reprinted with permission from [ref 65], copyright Optical Society of America. (F) Schematic of the variation of gyroscope output when a new six-state modulation method is applied. The

sequence of six phase states within one modulation cycle is: $\theta, 2\pi-\theta, -\theta, -\theta, -2\pi+\theta, \theta$. And its corresponding rotational speed demodulation sequence is {-1, +1, +1, -1, -1, +1}. (G) Schematic of the variation of gyroscope output when the new six-state modulation method is applied after considering the feedback link and the 2π reset error. The demodulation sequence for the 2π reset error is {+1, -1, +1, 0, -1, 0}. Copyright 2024 Optica Publishing Group. (H) Complete circuit schematic of the fully differential photodiode amplifier. Reprinted with permission from [ref 68], copyright Optical Society of America. (I) Schematics of detailed configuration of ISP-IFOGs and CW/CCW light path. Copyright 2021 Optical Society of America.

The Faraday effect is also an important source of bias errors in IFOGs. It mainly manifests as the phase drift of counter-propagating light waves in the fiber coil under the action of a time-varying radial magnetic field due to polarization non-reciprocity (PNR), which leads to the generation of bias errors in the system. It is worth noting that the effect is insensitive to the axial magnetic field. To suppress such errors, two main technical approaches are adopted from the perspective of device structure. Firstly, a magnetic shielding shell made of high-permeability metal materials [69] is employed to reduce external magnetic field interference through physical isolation. Secondly, a fiber coil is constructed using photonic crystal fiber (PCF), which leverages its unique waveguide structure to enhance resistance to magnetic field disturbances [70]. In addition, optical compensation techniques can also be applied to suppress such errors. It has been demonstrated that the Faraday effect-induced phase drift in two orthogonal polarization directions of PM fiber (PMF) has opposite polarities. Thus, combining the interference signals of both polarization directions can effectively suppress the phase drift induced by the Faraday effect [71-73]. An inverse symmetric polarization IFOG scheme (ISP-IFOG) can also suppress the bias error caused by the Faraday effect in the radial magnetic field (Fig. 5I) [74]. The method involves a novel fiber coil design that adjusts the angles of multiple fusion splices in the PMF, allowing light in two orthogonal polarization states to propagate within the coil. The scheme takes advantage of the opposite polarization characteristics of the light in the slow axis and the fast axis under the magnetic field, and the phase differences mutually cancel out. Since the total phase difference of the IFOG is the sum of the phases of the lights passing through the fiber coil on the slow axis and the fast axis, the phase drift caused by the Faraday effect can be reduced, thereby suppressing the bias error.

Furthermore, for the PNR issue in IFOGs, the polarization state control scheme is usually adopted. Specifically, the method retains only a single polarization state while fully suppressing the other. The implementation of the technique primarily depends on the application of high-birefringence PMFs. The core principle is to utilize the large propagation constant difference between the two orthogonal polarization modes, inherent to the high birefringence of the fiber, to effectively suppress polarization mode coupling during transmission. In addition, it has been demonstrated that the PNR phase shifts of the two orthogonal polarization modes supported in a dual-polarization IFOG have opposite signs [73,75-77]. When their intensities are balanced, these phase shifts can optically cancel each other. As a result, the dual-polarization scheme has become one of the main techniques for suppressing PNR at present.

2.3 Scale factor accuracy improvement technology

The scale factor represents the proportionality between variations in the IFOG output and the corresponding input changes. High-precision IFOGs have extremely strict requirements for the accuracy of the scale factor, with non-linear errors controlled at 5 ppm or an even more superior level. The errors related to the scale factor of IFOGs mainly include scale factor nonlinearity, scale factor asymmetry, and scale factor repeatability (stability). There are many factors that cause the above errors. Internal factors consist of the average wavelength drift of the light source, spectral asymmetry, instability of the optical path loss, inherent errors in the demodulation system, gain instability of the optical path or the circuit, modulator nonlinearity, and modulation coefficient drift of the modulator. External factors are mainly problems such as endogenous variations due to temperature and irradiation. Therefore, there are various methods to improve and enhance the scale factor in response to the effects of different factors. The section describes the techniques applied to improve the accuracy of the scale factor in recent years.

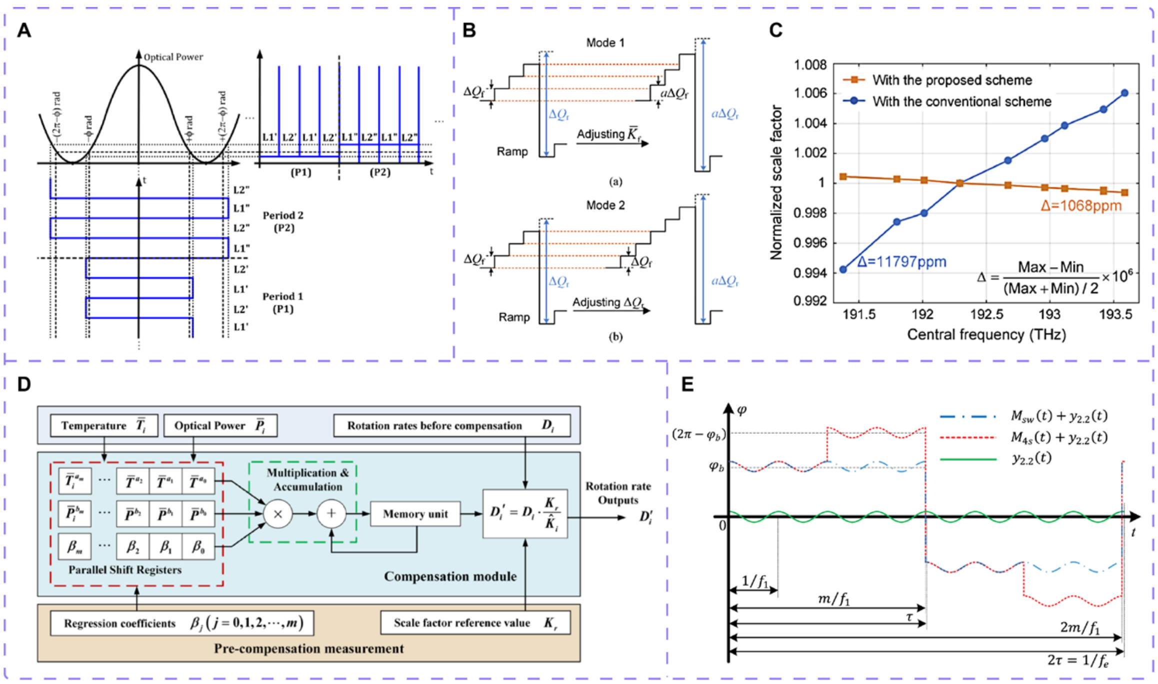

Fig. 6. Various schemes for scale factor accuracy improvement technology. (A) Phase shift induced by two-level and two-period modulation and gyroscope output signal when the modulation depth is not adjusted. Copyright 2018 Optical Society of America. (B) The actions on the ramp to null the reset-induced error by adjusting the feedback-chain gain (top panel) and the size of the digital-ramp register (bottom panel). (C) Test chart of the improvement effect on the scale factor performance of the scheme using the size of the digital-ramp register as the actuator. Reprinted with permission from [ref 86], copyright Optical Society of America. (D) FPGA compensation mapping structure for the multiple linear regression model with two parameters of optical power and temperature. The compensation method adopts a polynomial structure, utilizing only multiplication and addition operations, which facilitates its implementation on FPGA. Copyright 2024, IEEE. (E) Schematic of superimposing the even frequency doubled sine wave modulation on square-wave modulation and four-state modulation waveforms. The new modulation waveform induces output errors when the eigenfrequency varies, enabling real-time measurement of the eigenfrequency. Copyright 2024 Elsevier Ltd.

Currently, the digital signal processing system employed in IFOGs is the classical dual closed-loop feedback system [78], where the first closed-loop feedback system serves the purpose of conducting rotational measurements and the second closed-loop feedback system is used for the feedback gain control (also known as 2π control). The feedback gain error stands as one of the main influences on the linearity of the scale factor. Due to the temperature-induced drift of the modulation coefficient in the modulator, the modulation voltage changes, resulting in the feedback gain error in the IFOG output. After the feedback compensation of the second closed-loop system, the feedback gain error can be driven to converge to 0, thus eliminating its influence on the scale factor linearity. Meanwhile, combined with the four-state modulation technique, the rapid adjustment within twice the transition time is achieved, avoiding the time-consuming and inefficient scale factor control in the square-wave modulation technique. Similarly, Bacurau R M et al. proposed a two-level and two-period modulation technique [79] consisting of two intercalated periods of the square-wave modulation, one with a phase amplitude of $\pm\phi$ and the other with a phase amplitude of $\pm(2\pi-\phi)$, as shown in Fig. 6A. The angular velocity in the scheme is derived from the difference in the consecutive output levels, while the feedback gain error is obtained from the difference of the mean levels of the output from two consecutive periods. The scheme is similar to the four-state modulation technique and significantly improves the efficiency of the scale factor control while eliminating the feedback gain error. However, compared with the four-state modulation technique, the scheme allows for a slower ADC conversion rate. The four-state modulation technique requires each level to last for half of the transit time, which means the ADC conversion rate has to be twice that of square-wave modulation, increasing the risk of device resolution degradation. In contrast, the scheme does not have to meet such a requirement.

Various techniques have been developed to address the issues of average wavelength sensitivity and temperature sensitivity of the scale factor. Yang Y et al. established a simple wavelength-independent scale factor model [80], innovatively employing a compensation method based on intrinsic modulator parameters instead of the conventional MIOC half-wave voltage control. The approach achieved scale factor stability better than 10 ppm under extreme conditions (−40 °C to 60 °C, 100 krad (Si) radiation). The core of the method lies in accurately measuring the temperature/wavelength characteristics of the MIOC half-wave voltage, and dynamically compensating based on the direct proportional relationship between the modulator parameters and temperature. The requirement for average wavelength stability of the light source in IFOGs with high scale factor performance is significantly reduced. In 2018, She X et al. revealed through multiphysics simulation and experimental validation that temperature-induced misalignment between the MIOC and the fiber coupling structure is the primary factor of wavelength drift, which further leads to scale factor instability. Moreover, the temperature-induced misalignment is related to the initial alignment parameters of the coupling structure [81]. By optimizing the mode field of the lithium niobate ($LiNbO_3$) waveguide and precisely controlling the initial alignment, the approach effectively resolves the wavelength stability issue of the MIOC, thereby improving the scale factor stability of the IFOG. However, achieving optimal wavelength stability inevitably results in an increase in coupling loss. In the same year, Yang L et al. utilized the thermally induced wavelength drift characteristics of a positive temperature-dependent Er-doped superfluorescent fiber source (PT-EDSFS) to actively compensate for scale factor errors caused by the fiber coil and average wavelength variations [82], thereby meeting the requirements of high-precision IFOGs in practical applications. In 2020, Liu B et al. embedded a fiber temperature sensor inside the IFOG fiber coil, achieving accurate measurement of the thermal field within the coil. By incorporating input angular rate information, they developed a hybrid compensation model to compensate multi coefficient scale factor errors [83], effectively improving the performance of the IFOG scale factor. A twin-peaks light source scheme was also introduced for fast and accurate compensation of the IFOG scale factor [84]. The approach demonstrated that the temperature sensitivity of the dual-wavelength scale factor ratio is solely determined by the temperature characteristics of the gyroscope output ratio corresponding to the two wavelengths. Therefore, effective compensation of the scale factor can be achieved by establishing a temperature variation model of the dual-wavelength gyroscope output ratio. The scheme confirmed that the scale factor drift after compensation was suppressed by 30 times compared to that before compensation within the temperature range of -10~50 °C.

In addition, a multi-modulus temperature compensation model was also studied [85]. In the scheme, multiple precision temperature sensors are embedded in the fiber coil to construct a multi-modulus compensation model. Within the temperature range of 22.5-27.5 °C, the stability of the scale factor is improved to 1 ppm, which breaks through the precision limitation of traditional single-point temperature measurement and solves the hysteresis effect of the traditional IFOG scale factor model. In 2022, Zheng Y et al. eliminated the ramp-reset-induced errors by using the size of the digital-ramp register as an actuator in the closed-loop scheme instead of the conventionally-used feedback-chain gain (Fig. 6B), which significantly reduced the wavelength dependence of the scale factor (Fig. 6C) [86]. The scheme systematically studied the wavelength dependence of the IFOG scale factor. When the feedback gain is constrained by the closed-loop scheme to eliminate the ramp-reset-induced errors, there is a dependence relationship between the scale factor and the wavelength. However, when the actuator of the closed-loop scheme for the reset error is shifted from the feedback gain to the size of the digital-ramp register, the scale factor is almost independent of the wavelength. Therefore, the scheme can provide an effective reference for achieving better scale factor accuracy (less than 1 ppm) in high-precision IFOGs. It should be noted that the approach introduces a slight increase in the scale factor nonlinearity, asymmetry, and output noise of the gyroscope. In 2023, Jin J et al. constructed a multiple linear regression model (MLRM) with two parameters of optical power and temperature (Fig. 6D). Without adding any additional hardware devices, they improved the scale factor repeatability by 20 times [87]. Besides the above techniques, a series of approaches have been investigated to achieve better scale factor stability, including redesign and optimization of broadband light sources [82,88,89], measurement of wavelength distribution [90], and the introduction of photonic crystal fibers [91].

Spectral asymmetry and limited temporal coherence can also lead to the non-linearity of the scale factor. Wang X et al. investigated the influence of interference fringe visibility changes on closed-loop IFOGs operating across multiple fringes [92]. They conducted a detailed analysis of the scale factor non-linearity problems caused by spectral asymmetry and limited temporal coherence. The study demonstrated that spectral asymmetry can lead to significant scale factor non-linearity that varies with the fringe order, while limited temporal coherence does not result in notable non-linearity. By introducing a compensation model, the scale factor non-linearity was suppressed by 9.3-fold and 4.2-fold in two sets of gyroscope data, respectively. It is worth mentioning that the research team additionally proposed a scheme based on real-time feedback compensation of the proper frequency [93], which significantly improved the scale factor performance. Different from the techniques such as the temperature-induced parameter variation model for scale factor compensation, the scheme exploits the characteristic that the two factors (the length L and diameter D of the fiber coil) causing the instability of the scale factor also result in changes in the proper frequency of the IFOG. By employing the even frequency doubled sine wave modulation technique (Fig. 6E), the proper frequency can be measured in real time. A closed-loop control of the proper frequency is then established to compensate for variations in L and D, thereby achieving compensation for scale factor stability. The method can improve the stability of the scale factor by 6.7 times and offer significant potential and reference value for enhancing the scale factor accuracy of high-precision IFOGs.

Implementation of lasers in place of broadband light sources provides a novel approach to improving the scale factor stability of IFOGs. In 2015, Chamoun J. N. et al. investigated an analytical model of polarization coupled noise for lasers with arbitrary linewidths, and achieved a low-drift IFOG driven by a laser with a linewidth of ~100 MHz [94]. They also proposed that the laser light source, with highly stable average wavelength characteristics, can effectively solve the issue of poor scale factor stability of IFOGs utilizing incoherent light. In 2024, Chen X et al. replaced the traditional super fluorescent fiber source with a broadened laser, improving the scale factor stability of the closed-loop IFOG from 3.05 ppm to 0.38 ppm [95], which is one of the best scale factor performances achieved to date. Therefore, laser-driven IFOGs offer an option for further improving the scale factor performance in the future.

2.4. Some optimization technology

Besides the above-mentioned techniques for improving the accuracy performance, several optimization and auxiliary techniques also have important reference value for enhancing the accuracy performance of IFOG.

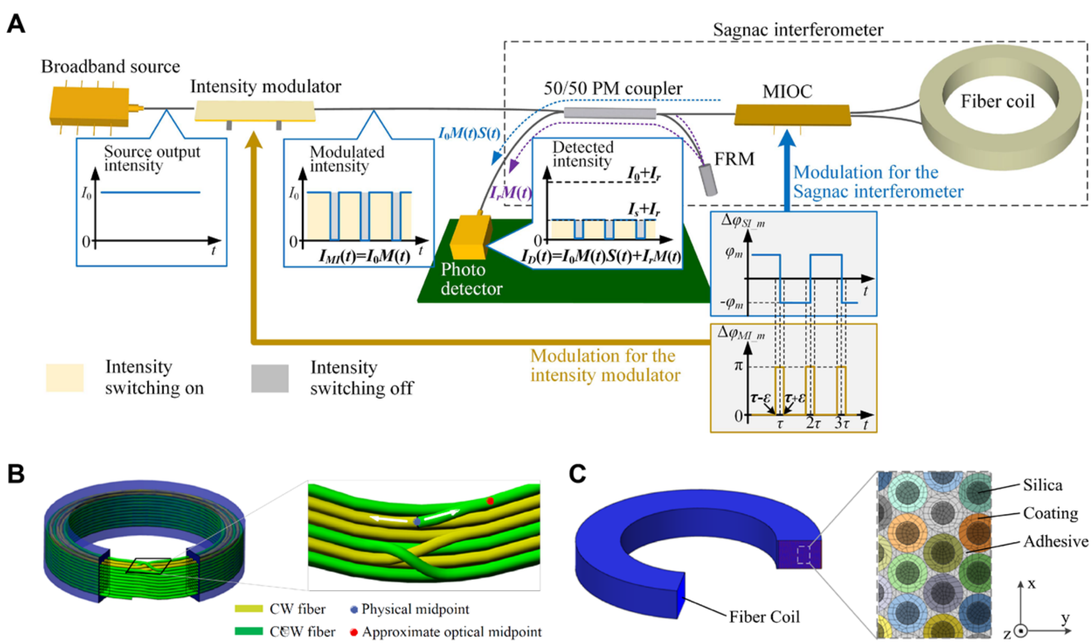

Fig. 7. Schemes for other optimization technology. (A) Schematic of the scheme for removing the intensity spikes using light-intensity switching. The phase difference of π at the intensity modulator leads to the intensity switching off where the spikes occur and the phase difference of 0 leads to the intensity switching on at other moments. Copyright 2024, IEEE. (B) Schematic of the symmetry problem of the fiber coil caused by the optical midpoint deviation. Differences in material properties cause variations in the refractive index, resulting in a shift of the optical center originally coincident with the physical midpoint of the fiber coil. (C) Schematic for the finite element model of the fiber coil. Copyright 2023 Optica Publishing Group.

The multiple harmonics modulation technique has achieved an ARW of $5.2\times10^{-6}\ °/\sqrt{h}$ and a bias instability of $10^{-5}\ °/h$ by optimizing the bias modulation of the IFOG [96]. In the scheme, multiple harmonics sine wave modulation is used, integrating the advantages of both square-wave and sine-wave modulation. It can produce a spike-free output while effectively reducing RIN and TPN. Similarly, the light-intensity switching technique based on a Mach-Zehnder-interferometer (MZI) can effectively address the issue of photodetector saturation caused by intensity spikes in IFOGs [97]. By inserting an MZI-based light intensity modulator at the light source output and synchronizing its phase modulation with the Sagnac interferometer (as shown in Fig. 7A), the light intensity can be actively switched off to eliminate predicted intensity spikes, while remaining open at other times to preserve valid signals. The method increases the detected light intensity to 3.57 times the conventional limit and reduces the ARW coefficient by 37.50%, significantly enhancing measurement accuracy. It offers a feasible approach for the miniaturization and environmental adaptability optimization of high-precision IFOGs. However, the intensity modulator also introduces new error sources such as insertion loss, polarization-dependent errors, and spectral distortion, leading to increased RIN and degraded scale factor performance.

To address the $1/f$ noise issue in the modulator driver circuit of IFOGs, Teng F et al. employed low $1/f$ noise operational amplifiers (such as the ADA4805), reducing the ARW coefficient introduced by the driver circuit to a low level ($1.6\times10^{-4}\ °/\sqrt{h}$) [98]. The study not only validated the accuracy of the proposed noise model but also provided a theoretical basis and engineering implementation approach for the noise optimization of IFOGs.

Temperature-induced errors and process challenges in high-precision fiber coil winding have always been the key limiting factors for improving the accuracy performance of IFOGs. Zheng Z et al. established a full-parameter mathematical model of the temperature errors of fiber coils. Through simulation, they optimized the winding scheme, material selection, and process parameters. They also combined a self-developed winding machine with an online detection and compensation system to correct errors such as stress and polarization crosstalk in real time [99]. The whole integrated technology covers material selecting, automatic winding, thermal treatment, and coil inspection, and has improved the precision of the fiber coils to 0.0005°/h. The method systematically addresses the temperature sensitivity and process complexity of long fiber coil winding, providing a reliable core component foundation for high-precision IFOGs.

The symmetry of the fiber coil (Fig. 7B) affects the temperature performance of IFOGs. Hu X et al. developed a temperature drift model including the Shupe effect and the T-point effect (mean temperature derivative), and proposed a compensation method based on fiber coil symmetry optimization using finite element simulation (Fig. 7C) and optical frequency domain reflectometry (OFDR) technology [100]. By precisely adjusting the lengths at both ends of the fiber coil and the optical center symmetry, the method effectively reduces residual phase differences, improving the IFOG bias stability by 18-23%. The scheme provides an engineering-feasible detection and compensation technique for suppressing temperature-induced errors in high-precision IFOGs, with potential for further accuracy improvement through optimized detection precision in the future.

2.5. Discussion

As mentioned above, the improvement of IFOG precision performance covers multiple technical aspects and involves various key techniques. The subsection summarizes these techniques in Table 1 to provide a clear overview and comparison.

Table 1. Comparison of techniques for improving IFOG precision performance

Field	Scheme	Effect or advantage	Ref.
Relative intensity noise suppression technology	Conventional circuit noise subtraction scheme	Easy to operate and capable of precise control.	[23-24]
	Optimized circuit noise subtraction scheme	Reduces the degradation of correlation between signals.	[25]
	SOA gain saturation scheme	Easy to integrate, wide bandwidth, and significant performance.	[26]
	External modulation with an intensity modulator	High-speed modulation with significant suppression performance.	[28]
	All-optical suppression scheme	Simple structure, easy to implement, excellent RIN suppression.	[29-33]
	Dual-polarization scheme	Effective and comprehensive RIN management method.	[34-36]
	Feedback controlled internal modulation	Acts on the light source to suppress low-frequency components of RIN.	[37]
Quantization noise suppression technology	ADCs and DACs with higher resolution	Improves quantization accuracy.	\
	FILC scheme	Simple, easy to implement.	[39]
	Dual-electrode pair MIOC scheme	High-resolution measurement, quantization noise suppression.	[40]
Phase noise suppression technology	Time-delay scheme	Capable of suppressing residual coherent phase error.	[41]
	Dual-light source scheme	Significant suppression of residual coherent phase errors.	[41]
	Drive voltage compensation scheme	Enhances the flatness of the interference signal.	[42]
	Higher-order frequency modulation scheme	Effectively suppresses discrete components in TPN.	[45]
	Source-sharing IFOG fusion scheme	Suppresses self-noise by 5.8 times and bias instability by 5.2 times.	[46]
Random noise suppression algorithm	ASST-SUKF	Achieves 100-fold error suppression in quiescent condition.	[52]
	WCARKF	High robustness, with several hundred times error suppression.	[53]
	AMA-RWE-DFAKF	100-fold random error suppression.	[54]
	IRWE-AKF	Short computation time with significant noise suppression.	[55]
	SSA-AND	Balances AND convergence speed with noise suppression.	[56]
	EMD-IMF	Prevents the loss of useful information, self-adaptive.	[57]
	MPE-CEEMDAN	Superior noise suppression performance.	[58]
	TVAKF		[59]

Field	Scheme	Effect or advantage	Ref.
Bias drift suppression technology	Circuit optimization scheme	Reduces electronic crosstalk.	[61]
	Random phase step technique	Completely eliminates electrical crosstalk errors from a statistical perspective.	[63]
	Random phase modulation technique		[64]
	Six-state phase modulation scheme	Completely eliminates crosstalk at a modulation depth of $2\pi/3$.	[65]
	Improved six-state phase modulation scheme	Eliminates fixed phase delay crosstalk at any modulation depth.	[66]
	Periodic polarity conversion scheme	Eliminates bias errors induced by crosstalk.	[67]
	Fully differential photodiode amplifier	Suppresses electrical crosstalk, with high gain and wide bandwidth.	[68]
	Magnetic shielding scheme	Physical isolation, simple and effective.	[69]
	Photonic crystal fiber scheme	The fiber coil insensitive to magnetic fields.	[70]
	Inverse symmetric polarization scheme	Phase differences between slow and fast axis cancel each other, low cost.	[74]
	High-birefringence PMF scheme	Maintains only a single polarization state.	\
	Dual-polarization scheme	Phase shifts can cancel each other.	[75]
Scale factor accuracy improvement technology	Second closed-loop feedback technique	Suppresses feedback gain errors.	[78]
	Two-level and two-period modulation	Reduces the speed requirements of the ADC.	[79]
	Modulator intrinsic parameter compensation	Reduces the dependence of IFOG on light source wavelength stability.	[80]
	Study on MIOC wavelength-temperature stability	Effectively addresses wavelength stability issues of the MIOC.	[81]
	PT-EDSFS wavelength drift compensation	Active compensation, significant effect, and easy to integrate.	[82]
	Twin-peaks light source scheme	Scale factor drift suppressed by 30 times.	[84]
	Multi-modulus temperature model	Resolves hysteresis effects in compensation.	[85]
	Closed-loop control actuator alteration	Reduces the scale factor's dependence on wavelength.	[86]
	Dual-parameter MLRM	Improves scale factor repeatability by 20 times.	[87]
	Eigenfrequency closed-loop scheme	Improves scale factor stability by 6.7 times.	[93]
	Laser driver scheme	Improves scale factor stability (0.38 ppm).	[95]

Field	Scheme	Effect or advantage	Ref.
Some optimization technology	Multi harmonic modulation technique	Suppresses RIN and TPN, reduces ARW, no spike issues.	[96]
	Optical power switching technique	No spike issues, prevents detector saturation, and reduces ARW.	[97]
	Fiber coil symmetry optimization scheme	Effectively suppresses residual phase differences.	[100]

3. System Component Performance Improvement Technology

Enhancing the performance of IFOG system components is a fundamental prerequisite for achieving high-precision measurements. Optimizing the performance of the light source directly affects the detection sensitivity and stability of the gyroscope. The improvement of the multi-functional integrated phase modulator can effectively suppress errors and improve the accuracy of closed-loop control. Meanwhile, enhancing the performance of the optical fiber can significantly reduce system noise and drift, thereby enhancing the long-term reliability. The coordinated optimization of these core components jointly promotes the breakthrough of the overall performance of IFOGs. thereby meeting the stringent requirements for high-precision inertial measurement in cutting-edge fields. Within the scope of our review, we investigated the techniques for improving the performance of system components used in IFOGs in recent years.

3.1 Light source performance improvement technology

The commonly used light sources in IFOGs are super luminescent diodes (SLD) and amplified spontaneous emission (ASE) sources. Both offer advantages such as broad spectra, low coherence, high stability, and long lifespan. Improving the performance of these two types of light sources is also an important part of IFOG research.

Yang J et al. investigated the wavelength temperature stability of SLD [101]. By developing a temperature simulation model for the SLD light source, they designed a temperature control scheme based on dynamic compensation of a thermistor. The approach precisely matches the thermistor's resistance-temperature characteristics with the temperature field distribution of the tube core, enabling automatic adjustment of the set temperature within the TEC control loop. The scheme can improve the temperature stability of the SLD wavelength by 62% and effectively suppress temperature drift. Nevertheless, the efficacy of the compensation approach is limited by the inherent non-idealities of the thermistor. In addition, to address the degradation of the IFOG bias stability and scale factor performance caused by power fluctuations in the SLD, Feng G et al. designed a SLD fluctuation compensation scheme based on real-time power feedback [102]. They compensated the gyroscope output signal by detecting the differential signal of the SLD power (Fig. 8A), effectively reducing the gyroscope output errors caused by SLD power fluctuations. In the scheme, a new modulation scheme was constructed. On the basis of square-wave modulation, a modulation with zero phase-shift was added to achieve the detection of optical power. The above-mentioned scheme can also be applied to ASE light sources.

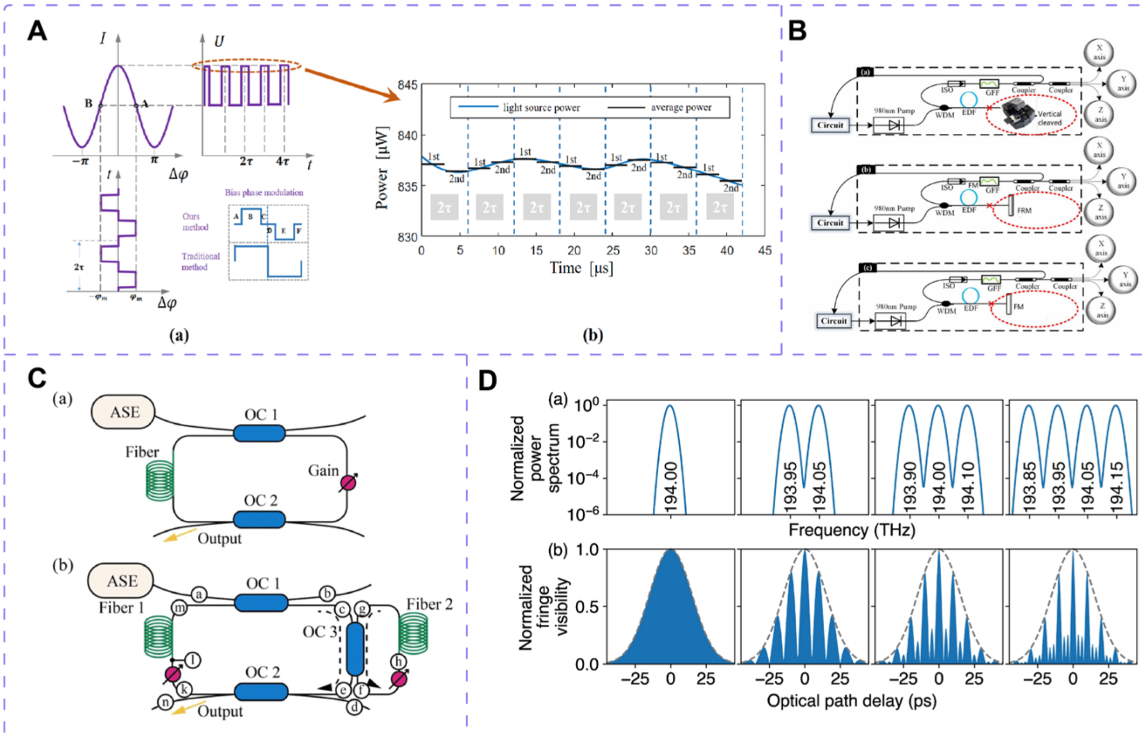

Fig. 8. Various schemes for light source performance improvement technology. (A) Schematics of the SLD power demodulation method and the SLD power signal. Copyright 2023, The Authors. (B) Schematics of three ED-SFS configurations based on vertically cleaved EDF pigtails, FRM and FM. Copyright 2020 Elsevier Inc. (C) Configuration of the D-FRR (bottom panel). The main FRR is formed via the resonator with nodes c -> d -> e-> l -> m-> c. The secondary FRR is formed via the resonator with nodes f -> h -> g -> f. Copyright 2023, The Authors. (D) Schematics of the normalized power spectrum and normalized fringe visibility of one through four lasers. Copyright 2024, IEEE.

In high-precision IFOG research, ASE light sources are commonly used due to their higher output power, superior average wavelength and power stability, and better fiber compatibility. In 2019, Chen X et al. proposed an optimized ASE light source scheme based on a dual-way backward configuration [103]. By optimizing erbium-doped fiber (EDF) parameters, designing a Gaussian spectral filter with a coherence function free of sub-peaks, and implementing power feedback control, the system achieved performance indicators of average wavelength variation less than 5 ppm and power fluctuation under 1% across the full temperature range. The output power exceeded 20 mW, effectively meeting the strict requirements of high-precision IFOG for the power stability and spectral symmetry of light sources. Furthermore, the team proposed an optimized broadband ASE light source scheme based on a two-stage EDF structure [104], effectively addressing the demands of ultra-high-precision IFOGs for source power, wavelength stability, and spectral width. The scheme employs a Faraday rotating mirror to suppress polarization-related noise, and uses a Gaussian spectral filter for full-spectrum shaping and filtering. Meanwhile, a high-stability temperature-controlled pump source is also utilized, making it an ideal light source for ultra-high-precision IFOGs. In 2020, Vostrikov E et al. conducted research on a wavelength stabilization scheme for EDF sources based on a dual-pass bidirectional pumping structure [105]. Through the jointly optimization of the pump ratio adjustment and a temperature-compensated filter, the approach effectively suppressed central wavelength thermal drift in high-precision IFOGs. In the scheme, the randomly selected length of the unpumped filter

EDF constrains further advancement of its ultimate performance. In the same year, Yang L et al. presented a method for low-cost vertically cleaved EDF pigtail based on Fresnel reflection [106]. They achieved a thermally-stable erbium-doped superfluorescent fiber source (ED-SFS) in one-stage backward configuration. Compared to conventional structures using Faraday rotating mirror or fiber mirror, the method significantly improved the thermal stability of the average wavelength of the light source and enhanced the scale factor performance of the IFOG (Fig. 8B). Moreover, Chen X et al. studied a novel method for suppressing the RIN of ASE light source based on a passive/active dual fiber ring resonator (D-FRR) [107]. They utilized the D-FRR as an optimized gain point filter, effectively overcoming the technical limitation of the traditional noise suppression scheme based on FRR [108], which requires matching the length of the sensing coil. At the same time, an equivalent RIN suppression was maintained (Fig. 8C).

A low-coherence light source scheme based on coordinated spectral broadening of four laser diodes via phase modulation has also been investigated (Fig. 8D) [109], achieving optimal performance for laser-driven IFOGs. In the scheme, the four lasers are simultaneously broadened using phase modulator driven by the same noise. Combined with optical gating technology to suppress the residual drift caused by the Kerr effect, the average wavelength stability of strategic-level IFOGs is achieved. The scheme provides a valuable reference for enhancing the performance of high-precision IFOG light sources.

3.2 Multifunction integrated phase modulator performance improvement technology

The MIOC refers to a Y-type $LiNbO_3$ waveguide with input and output pigtails, which integrates optical fiber splitter and phase modulator, as shown in Fig. 9A. $LiNbO_3$ materials possess a high electro-optic coefficient, a broad transparency window, and stable material properties. MIOCs based on $LiNbO_3$ fabricated by annealed proton exchange (APE) method exhibit single-polarization characteristics and can achieve very high polarization extinction ratios, making them one of the key components in current high-precision IFOGs [110]. Critical parameters of the MIOC, such as half-wave voltage, residual intensity modulation (RIM), and extinction ratio, directly affect the bias stability and scale factor stability of high-precision IFOGs. Therefore, performance enhancement of the MIOC is essential for realizing high-precision IFOGs.

To address the issue of parasitic refractive index drift in $LiNbO_3$ electro-optic modulators caused by surface layer charge relaxation, Pogorelaya D. A. et al. proposed a compensation method based on inverse filter pre-emphasis [111]. They used the filter to describe the inertial process of electro-optic modulation and suppressed the additional phase shift caused by the parasitic refractive index drift. In 2019, Miao L. et al. studied the frequency dependence of the half-wave voltage of the MIOC [112], and proposed a general measurement method based on phase-locked amplification and sinusoidal modulation. At the same time, they also investigated the interference intensity oscillation in IFOGs caused by the frequency-dependent half-wave voltage. The study offers a new error analysis direction for improving MIOCs in high-precision IFOGs.

Regarding the RIM in MIOCs, Zheng Y. et al. proposed a waveguide anti-reflection layer design that combines rough treatment and absorption coating [113]. The approach targets the nonlinear RIM issue caused by TE mode stray light being recoupled into the output fiber due to reflections from the waveguide bottom surface (Fig. 9B and C). By constructing an anti-reflection layer to attenuate stray reflections from the bottom facet of the waveguide substrate (Fig. 9D and E), the nonlinear RIM was reduced by an order of magnitude. Therefore, the technique effectively suppresses phase noise and bias drift in high-precision IFOGs. Additionally, Shulepov V. A. et al. proposed a method to suppress RIM by optimizing the geometry of diffused-channel Ti:$LiNbO_3$ waveguides [114]. Their design introduces specific angular bends at the input/output ends of the MIOC to block parasitic propagation paths through the substrate, thereby reducing the RIM value of the MIOC. The scheme imposes higher requirements on manufacturing precision.

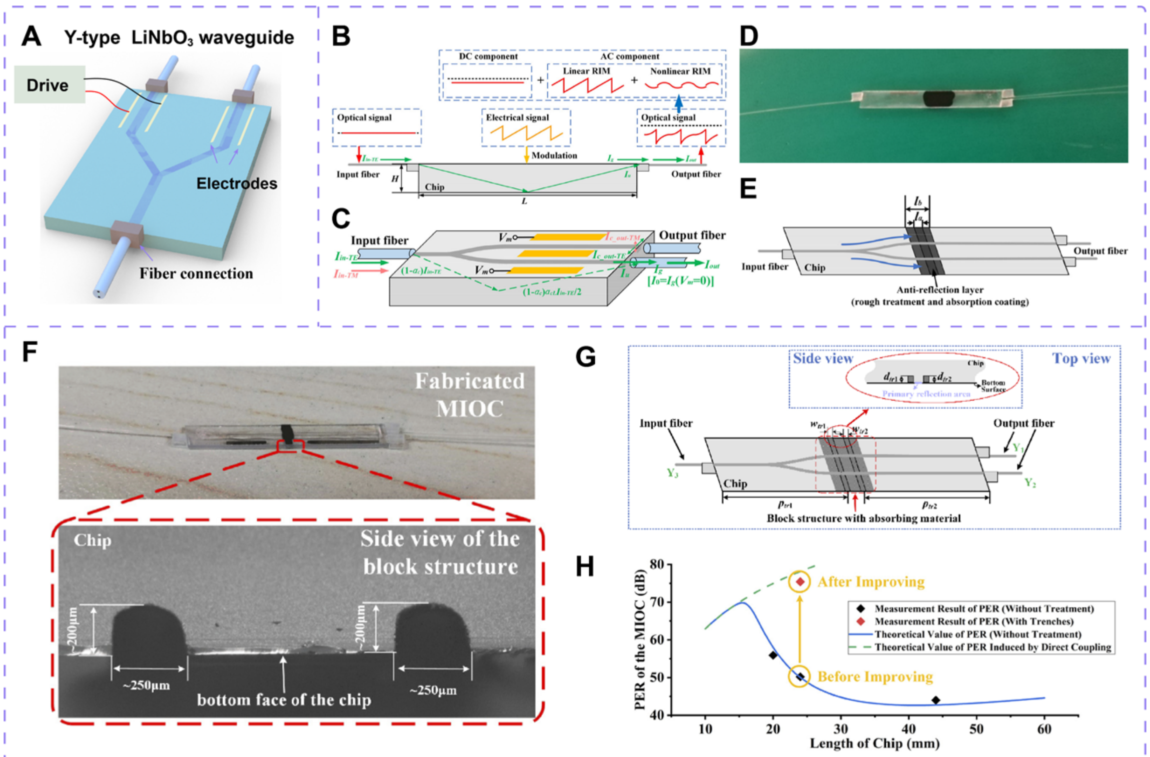

Fig. 9. Various schemes for MIOC performance improvement technology. (A) Structure of an MIOC. (B) Schematic of the output intensity of the MIOC with modulation applied. (C) Schematic of the light intensity variables in MIOC. The output signal contains both linear and nonlinear RIM components. (D) The product of the MIOC with the anti-reflection layer. (E) Schematic of the optimization scheme for the anti-reflection layer at the bottom of the MIOC. The surface of the anti-reflection layer must cover the reflection region. Copyright 2020, IEEE. (F) The product of the MIOC with the block structures at the bottom. (G) Schematic of the optimization scheme for the block structures on the bottom of the MIOC. The block structures are located on both sides of the central main reflection region. (H) Comparison of measured and theoretical PERs of MIOC with different chip lengths, and improvement of the PER. Reprinted with permission from [ref 115], copyright Optical Society of America.

Furthermore, Gao F. et al. proposed the double absorption trenches structure for the MIOC to enhance the polarization extinction ratio (PER), as shown in Fig. 9F [115]. The study investigated PER degradation caused by reflection-induced recoupling of waveguide unguided TM modes in the substrate (Fig. 9G), and introduced double absorption trenches at the bottom of the substrate to effectively block the reflection-induced recoupling paths of the TM mode. The design improved the average PER by approximately 25 dB, achieving the value of the PRE exceeding 75 dB (Fig. 9H). The structure is simple and easy to fabricate, avoiding the complex processing required by the traditional cut-off-bonded method [116], and offers a practical solution for enhancing the performance of high-precision IFOGs.

In addition, the electrostrictive effect of the MIOC crystal can induce phase errors in the gyroscope signal modulation, leading to output signal oscillation and further generating quantization noise. Shu X. et al. investigated a modulation signal frequency optimization scheme [117], in which the frequency components of the IFOG modulation square-wave are adjusted to avoid the resonant frequency band of the MIOC, thereby effectively decoupling the IFOG output from the crystal's electrostrictive response. By optimizing the fiber coil length to control the modulation square-wave's frequency composition, the scheme significantly reduced the output fluctuation amplitude of the IFOG, which has high engineering practicability and cost-effectiveness.

Optimization of the fabrication process of $LiNbO_3$ crystals is also an effective approach to improve MIOC performance. Karagöz E. et al. carried out pre-annealing treatment on $LiNbO_3$ wafers at 500 °C, which effectively reduced lattice defects and charge traps. As a result, the optical performance and temperature stability of the MIOC are significantly improved [118].

3.3 Fiber performance improvement technology

The improvement of fiber performance plays a decisive role in the the precision and environmental adaptability of IFOGs. Traditional fibers, limited by material impurities and types, exhibit optical characteristics that are susceptible to temperature fluctuations, magnetic fields and radiation, resulting in the degradation of IFOG performance. In recent years, numerous studies have introduced special fibers into IFOG systems to address these limitations, primarily including PCFs and multi-core fibers (MCFs).

PCF is a type of optical fiber characterized by a unique microstructure, where its cross-section exhibits a regular two-dimensional periodic arrangement: a high-refractive-index silica glass serves as the host material, embedded with a periodic array of low-refractive-index air holes. The exquisite microstructure enables PCF to guide light through photonic bandgap effect or total internal reflection mechanism, resulting in optical properties distinctly different from conventional fibers. PCFs mainly include solid-core PCF (SC-PCF) (Fig. 10A) [119,120] and hollow-core PCF (HC-PCF) [121,122] (Fig. 10B). SC-PCF relies on the principle of total internal reflection, while HC-PCF primarily operates based on the photonic bandgap effect. The special structure of PCF makes it significantly less sensitive to environmental factors such as temperature, magnetic fields, and radiation compared to traditional fibers. Therefore, PCF has become a key research focus for the improvement of IFOG performance.

In 2017, Sun Z et al. studied the bias thermal stability of IFOG based on polarization-maintaining PCF (PM-PCF) [123]. Compared to conventional PM-fiber (PMF), PM-PCF exhibits lower thermal sensitivity of birefringence, which effectively enhances the thermal stability of IFOG. However, the PM-PCF used in the study exhibits severe backscattering and very high transmission loss. In 2020, to solve the nonreciprocity problem caused by higher order modes transmitted in previous PM-PCF, an optimized small coating single mode PM-PCF scheme was proposed [124]. Using the full vector finite element method, the coating and cladding diameters were optimized to significantly suppress higher order modes. Experimental verification in IFOG systems demonstrated improved static and full-temperature bias stability. The scheme is suitable for high-precision IFOG. In the same year, Yang B et al. reported a novel PCF coil with a length of 3800 meters implemented using the octupole symmetric winding structure, achieving a IFOG bias stability of 0.001°/h [125]. The scheme has been verified on orbit by the SJ-20 satellite. After continuous operation in the geostationary orbit for two months, there was no degradation of accuracy parameters, which confirmed the adaptability of IFOG based on PCF to the space environment. In 2021, Teng F et al. established a comprehensive model of the ARW coefficient that includes the optical fundamental noise, the electronic circuit noise and the backscattering and back-reflection noise induced by HC-PCF. Moreover, they adopted the optimal modulation phase parameter scheme, which reduced the ARW coefficient by 17%, thereby significantly enhancing the accuracy of the IFOG [126]. In the same year, Kong L et al., building upon their earlier research [127], further optimized the model and fiber drawing process (Fig. 10C to E), successfully reducing the transmission loss of PCF to 1.1 dB/km [128]. Based on the optimized small diameter and low loss PCF, the team developed a new generation of photonic crystal IFOG products, achieving a static bias stability better than 1×10^{-3} °/h and an angular random walk (ARW) below 1.7×10^{-4} °/$\sqrt{h}$. These products have laid a solid foundation for the practical application of PCF gyroscopes in the aerospace field. In 2023, Liao M et al. systematically analyzed the effects of core, core-side holes, and diameter defects on the optical performance of HC-PCF, which is prone to structural defects during fabrication. They proposed solutions such as optimizing the microstructure and coating, effectively reducing fiber loss and thermal sensitivity [129]. The achievement not only provides a key theoretical reference for the fabrication of long-distance high-performance HC-PCF, but also improves the temperature stability and accuracy of IFOG, promoting the practical application of HC-PCF in the field of high-precision inertial navigation.

The MCF technology is primarily applied to enhance the transmission capacity of optical networks [130]. To address the challenge of balancing miniaturization and long optical path in traditional IFOGs, Mitani S et al. proposed an innovative structure based on MCF and fan-in/fan-out (FIFO) devices [131]. By using a 102-meter seven-core MCF coil, they achieved an equivalent long optical path and verified the feasibility of the MCF-based scheme in an open-loop IFOG for the first time, as shown in Fig. 10F. The approach leverages the core number scalability of MCF to overcome the physical length limitations of fiber coils, making 10 km level equivalent optical paths possible in the future. The research blazes a new trail for high-precision miniature IFOG. MCF-based fiber coils can also address the industry pain points of low efficiency and high cost associated with manual winding of traditional IFOG coils. Nonetheless, the large size of the FIFO device conflicts with the goal of miniaturization. In addition, this scheme involves a potential risk of core-channel crosstalk, and these limitations are expected to be resolved through process optimization in future work. Smith R H et al. broke through the automation bottleneck of coil winding using the MCF technology (Fig. 10G) and the dual axis symmetric (DAS) winding pattern [132]. In addition, a novel IFOG scheme based on dual-core fiber (DCF) was also proposed [133]. By achieving stationary phase shifts with opposite signs in different cores of the DCF, differential rotation rate measurement is realized, significantly improving measurement sensitivity and SNR. The scheme adopts an all-fiber, mechanical-free phase shifter design, featuring vibration resistance, overload resistance, and low power consumption, offering a new approach for high-precision IFOGs. The scheme can only compensate for the common-mode noise affecting both cores, while the non-common-mode noise limits further enhancement of system performance.

Li X et al. also improved traditional PMF by proposing two types of innovative hybrid panda-type PMF structures based on the collaboration of geometry and stress effects (Fig. 10H) [134]. The geometric anisotropy was enhanced through the horizontal-elliptical core PMF (HE-PMF) structure. Combined with fiber coil winding techniques, they achieved a static extinction ratio of over 30 dB. The study also revealed that the longitudinal-elliptical core PMF (LE-PMF) suffers performance degradation due to geometric birefringence suppression effects. The work established the optimal technical route of HE-PMF combined with multipole winding for high-precision navigation systems, promoting the practical application of high-precision IFOGs in harsh environments such as aerospace.

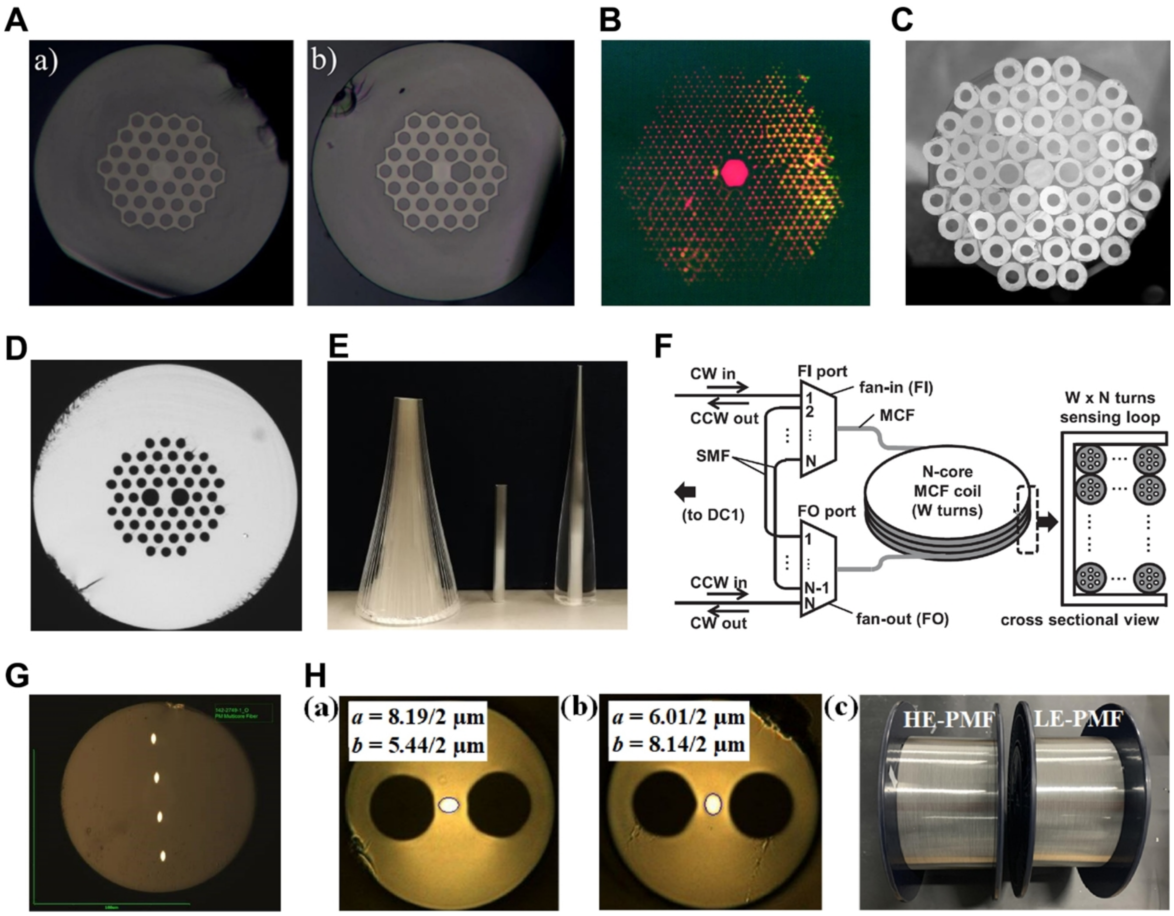

Fig. 10. Various schemes for fiber performance improvement technology. (A) The microscopic image of SC-PCF. Copyright 2015, The Authors. (B) The optical micrograph of HC-PCF. Copyright 1999, The American Association for the Advancement of Science. (C) The stacked preform of the PCF. (D) The cross section of the PCF. (E) The cones drawn from preforms of the PCF. Copyright 2021, IEEE. (F) Schematic of the fiber coil using MCF with bundled N-core waveguides. Copyright 2019, IEEE. (G) End view of Corning's PM 4 core fiber. Copyright 2024, SPIE. (H) Cross-sectional SEM images and 1-km fiber products for the HE-PMF and LE-PMF. Copyright 2023 Optica Publishing Group under the terms of the Optica Open Access Publishing Agreement.

3.4 Discussion

The optimization of system component performance serves as the fundamental foundation for enhancing IFOG performance. For clarity and comparison, these optimization techniques are summarized in Table 2.

Table 2 Comparison of techniques for improving the performance of IFOG system components

Field	Scheme	Effect or advantage	Ref.
Light source performance improvement technology	Thermistor-compensated SLD temperature control.	SLD wavelength temperature stability improved by 62%.	[101]
	Real-time power feedback compensation scheme	Real-time power monitoring achieved through modulation.	[102]
	Dual-way backward configuration scheme	The average wavelength change (full temperature): <5 ppm.	[103]
	Two-stage EDF structure scheme	Power output above 30 mW, high wavelength stability.	[104]
	Dual-pass bidirectional pumping scheme	The central wavelength stability for 2 h is 1.3 ppm.	[105]
	Low-cost vertically cleaved EDF pigtail scheme	Enhances the thermal stability of the average wavelength.	[106]
	RIN suppression scheme based on D-FRR structures	Supports both passive and active configurations.	[107]
	Multi-laser coordinated broadening scheme	Average wavelength stability better than 0.5 ppm.	[109]
MIOC performance improvement technology	Compensation based on inverse filter pre-emphasis	Suppresses phase shifts induced by parasitic refractive index drift.	[111]
	Waveguide anti-reflection layer design scheme	Nonlinear RIM reduced by an order of magnitude.	[113]
	Geometrical optimization for Ti:$LiNbO_3$ waveguides	Effectively suppresses residual intensity modulation.	[114]
	Double absorption trenches structure scheme	PER improved by approximately 25 dB.	[115]
	Modulation frequency optimization scheme	Avoids effects induced by electrostrictive.	[117]
	Pre-annealing treatment scheme	Effectively reduces lattice defects and charge traps.	[118]
Fiber performance improvement technology	PM-PCF scheme	Enhances the thermal stability.	[123]
	Small coating PM-PCF	Suppresses higher order modes.	[124]
	Novel PCF coil	Bias stability reaches 0.001°/h.	[125]
	Small-diameter, low-loss PCF scheme	Bias stability better than 0.001°/h.	[128]
	Study on structural defects in HC-PCF	Reduces fiber loss and thermal sensitivity.	[129]
	Scheme based on MCF and FIFO device	Breaks the physical length limitation of the fiber coil.	[131]
	Scheme based on MCF and DAS	Overcomes automation bottlenecks in winding processes.	[132]
	Scheme based on DCF	Accuracy improved ten times.	[133]
	Hybrid panda-type PMF structures	HE-PMF can achieve a static extinction ratio of over 30 dB.	[134]

4. Environmental Adaptation Improvement Technology

The improvement techniques for environmental adaptability of IFOGs directly determine their reliability and accuracy in complex application scenarios. External factors such as temperature and magnetic field can directly affect the geometric, physical, and even chemical properties of system components, thereby degrading the output performance of IFOGs. Many of the techniques mentioned above have already demonstrated conspicuous improvements in the environmental adaptability of IFOGs. The section further supplements recent advancements in adaptability techniques concerning temperature and magnetic field.

4.1 Temperature Adaptation Improvement Technology

The temperature adaptability technology of high-precision IFOGs aims to reduce the temperature sensitivity of IFOG performance. Zhang Y et al. proposed a thermal error compensation method for IFOGs based on temperature diffusion theory [135]. By establishing a linear model between nonreciprocal phase-shift and asymmetric fiber tail length, they revealed the mechanism of phase error generation under thermal disturbances. The scheme achieves active compensation of thermal-induced phase-shift within the temperature range of -40°C to 60°C through precise control of fiber tail length symmetry, effectively improving the environmental adaptability of IFOGs. Wang X et al. proposed a fiber coil optimization scheme based on a thermal stress induced Shupe error model [136]. By means of analyzing the temperature characteristics of the Young's modulus and thermal expansion coefficient of coating adhesives, they investigated the influence of the thermal stress distribution in the fiber coils caused by these factors on the nonreciprocal error of the gyroscope. The scheme established three principles for coating material selection: high glass transition temperature, linear variation of Young's modulus, and linear variation of thermal expansion coefficient. The study provides theoretical guidance and process standards for enhancing the stability of high-precision IFOGs in complex temperature environments. Li X et al. demonstrated that using the elliptical-core panda PMF with high birefringence and low effective mode refractive index can greatly improve the temperature stability of IFOGs [137]. Considering both performance and cost, the elliptical-core panda PMF is an excellent choice for enhancing the temperature adaptability of IFOGs.

In addition to the hardware optimization designs mentioned above, current IFOG temperature error compensation techniques mainly focus on software approaches due to their high compensation flexibility. Traditional compensation schemes typically establish polynomial models between the IFOG output and temperature-dependent terms, then use methods such as least squares and regression analysis for fitting. Further compensation of the output is carried out to achieve temperature insensitivity. However, traditional compensation methods cannot adapt well to temperature variations in complex environments. Therefore, artificial intelligence (AI) methods have been introduced into IFOG temperature compensation. AI methods can effectively capture the nonlinear characteristics of data by improving sample construction, designing complex network architectures, and optimizing training and testing processes, showing great potential in the field of IFOG software compensation. The core idea is to use temperature-related parameters as input samples and gyroscope output as labels, training models to accurately predict gyroscope output from temperature information. Many AI-based schemes have been realized for compensating IFOG temperature errors [138-140]. Additionally, Chen G et al. proposed a multivariate temperature field drift model based on variational mode decomposition (VMD) and particle swarm optimization (PSO) to address the drift problem of large-diameter high-precision IFOGs under complex temperature fields, and employed Support Vector Regression (SVR) to train the temperature drift model [141]. The approach utilizes VMD to accurately extract temperature drift signal features and combines PSO to optimize compensation parameters. As a result, it achieves effective temperature drift compensation for large-diameter IFOGs. The compensation accuracy is improved by at least 60% compared with other typical schemes, providing a reliable temperature anti-interference solution for high-precision navigation systems. In the approach, the training data are required to include full-temperature-cycle measurements at different rates of temperature change, with each rate maintained for several hours to ensure stability, imposing stringent requirements on the training dataset.

Further, Liang S et al. proposed a temperature sensor-independent IFOG temperature error compensation method based on a long short-term memory (LSTM) network in response to the problem of AI schemes relying on temperature information [142]. The method constructs time series samples solely from historical IFOG output data to train the model that predict future output, thereby indirectly learning the correlation between temperature variations and gyroscope errors. Therefore, it can effectively eliminate the need for additional temperature sensors. In the scheme, the network structure consists of five LSTM units and one fully connected layer. The training data must cover the full temperature range from −40 °C to 60 °C and include three trends: heating, holding, and cooling. Compared with traditional compensation methods that rely on temperature information, the method not only improves the accuracy of the IFOG (reducing bias instability by 89.57% and ARW coefficient by 22.53%) but also reduces system complexity and saves computing resources. It is especially suitable for static or low-rotation scenarios such as satellite observation and also provides important reference for compensating other errors (e.g., scale factor error) in dynamic applications.

The aforementioned temperature adaptation improvement technology has different emphases depending on the application scenario. For instance, the LSTM-based scheme without temperature sensors eliminates the need for temperature sensors and associated circuitry, simplifying the IFOG hardware and reducing the failure rate, making it particularly suitable for aerospace applications that require high reliability and low fault rates. In contrast, for applications such as high-precision terrestrial navigation and deep-sea exploration, which can provide stable power and computational resources and involve relatively complex temperature variation patterns, the multivariate temperature field drift compensation scheme offers greater advantages. In cost-sensitive civilian and industrial fields, optimizing the fiber coil and the material itself represents a reliable approach to ensuring basic performance while controlling costs. By aligning compensation techniques with specific application scenarios, the most effective temperature adaptation solutions for IFOGs can be provided across different domains.

4.2 Magnetic Field Adaptation Improvement Technology

The magnetic field adaptation improvement technology of high-precision IFOGs mainly focuses on magnetic shielding optimization of the optical path and suppression of magnetic-induced errors, which has been discussed in the section of "Bias drift suppression technology". This section provides a supplement on the magnetic field adaptation techniques for IFOGs. Research on IFOG magnetic field errors began in the 1980s, and numerous studies have established a comprehensive theoretical framework [143-151], providing theoretical guidance for improving the magnetic field adaptation of IFOGs.

Fiber twisting increases the proportion of elliptical polarization in the fiber, leading to Faraday phase errors induced by magnetic fields [145]. In 1994, Litton Company investigated the addition of oppositely twisted fiber coils to compensate for IFOG errors caused by radial and axial magnetic fields, and experiments validated the feasibility of the approach [152]. Therefore, fiber untwisting is an effective method for enhancing magnetic field adaptation. In theory, fiber untwisting can be implemented using an untwisting device prior to the fiber winding process. However, in practical applications, the feasibility of using such devices is limited and the operation is challenging, mainly because the external twist of each fiber section in different layers does not vary uniformly.

Expanding on the foundation of [152], Honeywell Company connected multiple fiber coils with specific magnetic sensitivities and orientations (magnetic compensators) to the original fiber coil [153]. By adjusting their relative spatial orientations, the combined magnetic sensitivity vectors of these coils sum to precisely cancel the magnetic field errors of the original fiber coil. The approach no longer requires each individual compensator to exactly match a specific (axial or radial) component of the original fiber coil.

Magnetic shielding is a direct and effective method for improving the magnetic field adaptation of IFOGs. In 2007, Litton Company introduced a magnetic shielding structure characterized by high rigidity, reusability, and ease of disassembly and installation [154]. Furthermore, Li H et al. proposed an innovative design scheme of magnetic shield with annular cavity structure [155]. Through finite element simulation and experimental validation, they systematically studied the effects of factors such as the number of layers, the layer spacing, the via holes and the gap width on the shielding effectiveness. By adopting measures including the two-layer staggered structure, increased joint surface overlap width, installation of magnetic liner in the gap, and optimized arrangement of fastening screws, they effectively suppressed the impact of structural defects on shielding performance, providing a reliable solution for suppressing magnetic interference in high-precision IFOGs.

In addition, photonic crystal fiber has emerged as a fundamental solution to the magnetic field adaptability problem in IFOGs due to its extremely low sensitivity to the magnetic field. In 2017, Xu X. et al. measured the Verdet constant of PM-PCF using a white-light interferometer [156]. The experimental results showed that the Verdet constant of the PM-PCF is approximately 124 times lower than that of panda-type PMF. Theoretically, replacing panda-type PMF with PM-PCF could reduce the magnetic-field-induced error in IFOGs by approximately 25 times. The approach provides important guidance for the application of PM-PCFs in resisting magnetic interference.

The choice of techniques for improving magnetic field adaptation closely depends on the magnetic environment and system constraints of the target application. In industrial applications involving strong magnetic fields or maritime navigation near ship propulsion systems, direct and effective magnetic shielding is often the preferred solution, despite the associated increases in weight and cost. For aerospace applications, where weight and space are extremely limited, PCF-based approaches offer a better balance between high performance and miniaturization. In summary, magnetic field adaptation technology for IFOGs is evolving toward diversification, allowing the most suitable technique to be selected according to the specific application scenario.

4.3 Discussion

As mentioned above, technology for improving the environmental adaptability of IFOGs involves both temperature and magnetic field aspects, with various techniques aimed at improving the reliability and precision of IFOGs under complex operating conditions. A comparison of these techniques is presented in Table 3.

Table 3 Comparison of techniques for improving the environmental adaptation of IFOG

Field	Scheme	Effect or advantage	Ref.
Temperature adaptation improvement technology	Compensation method based on diffusion theory	Active compensation of thermal-induced phase-shift.	[135]
	Optimization scheme based on a thermal stress model	Establishes coating selection principles.	[136]
	Elliptical-core panda PMF	Improves temperature stability.	[137]
	Traditional polynomial compensation scheme	Simple and easily implement.	\
	Multivariate temperature field drift model	Improves accuracy by at least 60%.	[141]
	Sensor-independent compensation method	Reduces bias instability by 89.57%.	[142]
Magnetic field adaptation improvement technology	Fiber untwisting scheme	Suppresses Faraday phase errors.	\
	Oppositely twisted fiber coils scheme	Radial and axial magnetic field compensation.	[152]
	Multiple magnetic field compensators scheme	No precise match required.	[153]
	Magnetic shielding with annular cavity structure	shielding effectiveness reach 55 dB (radial) or 80 dB (axial).	[155]
	PCF-based scheme	Lower nonreciprocal error.	[156]

5. Conclusion

The development of fiber optic gyroscope technology benefits from the collaborative advancements in optical fiber technology, optical communication, optical sensing, microelectronics, signal processing, materials science, precision manufacturing, and related fields. Up to this day, FOG has achieved extreme-precision angular rate measurement, with long-term bias stability reaching the order of 10^{-7}°/h. The performance of high-precision IFOG is gradually approaching and even matching that of electrostatic gyroscope, with the potential to replace electrostatic gyroscope as the preferred choice in long-endurance inertial navigation system in the future. Moreover, with continuous improvements and optimizations in IFOG performance, the application fields have expanded into ultra-high-precision detection areas such as UT1 measurement, seismology and micro-angular vibration measurement. This review summarizes recent research progress on high-precision IFOG, focusing on technology for improving precision performance, system component performance, and environmental adaptability.

Although the performance indicators of IFOG have been significantly improved through the iteration of numerous methods such as material optimization, structural design, and signal processing, gyroscopes that can be engineered still have difficulty meeting the requirements in certain extreme environmental scenarios. The following are the issues and future improvement directions.

First, the precision limit of IFOGs is constrained by various noise sources. Among them, shot noise, arising from the quantization of the electromagnetic field, is the fundamental factor limiting the ultimate performance of IFOGs. Within the framework of classical techniques, the noise limit is insurmountable. However, novel quantum-enhanced interferometric sensing mechanisms utilizing squeezed light or entangled photon pairs can surpass the classical noise limit, making them essential approaches for further exploring breakthroughs in the precision limit of IFOG. Consequently, the construction and application of a stable "Quantum Fiber Optic Gyroscope" will be a critical research direction for high-precision IFOGs.

Second, increasing the fiber length L and coil diameter D to enhance the phase response is a common method for improving IFOG precision. However, the method inevitably leads to increased volume and mass, which is particularly problematic in applications such as microsatellites where strict constraints on size and weight exist. As a result, the development of compact, high-precision IFOG techniques has evolved into a critical direction. Ultra-thin fiber technology and multi-core fiber technology have emerged as promising candidate solutions.

Third, the optimization and compensation of multiple parameters and physical fields in IFOGs are mutually restrictive, leading to suboptimal improvements in performance. For example, increasing the size of the fiber coil introduces more complex temperature field variations. During spacecraft orbital transfer or deep-sea cruising, dynamic temperature fields and mechanical stress fields become intricately intertwined. Traditional single-point temperature compensation or single-parameter compensation techniques are inadequate or even ineffective, thus further degrading IFOG performance. Balancing the optimization ranges of various parameters and developing multi-physics collaborative modeling and compensation methods that integrate environmental, mechanical, and optical coupling effects represent a noteworthy research direction.

Fourth, as IFOG applications expand, the interactions and nonlinear characteristics of multiple environmental variables make it difficult to establish effective compensation models. These models must also exhibit high flexibility and adaptability. For instance, IFOGs must withstand extreme conditions such as high temperature, high humidity, and high pressure in deep-sea applications, while they are exposed to strong radiation and magnetic fields in space. Developing effective compensation models with self-adaptive and self-supervised learning capabilities for these extreme environments represents a critical future research focus for IFOGs. With the rapid development of machine learning and neural networks, these emerging tools should be promptly introduced to continuously overcome challenges such as environmental adaptability.

Fifth, the performance of high-precision IFOGs fundamentally depends on the quality of their system components. For instance, optical path loss limits the improvement of SNR, and spectral asymmetry leads to nonlinearities in the scale factor. Therefore, it is necessary to further explore system components such as light sources with broad bandwidth, temporal stability, low coherence, and high output power; fiber coils with better environmental adaptability and superior optical transmission properties; and MIOCs with higher polarization extinction ratio. Raman-amplified broadband sources pumped by tunable random fiber lasers [157], as well as specialty fibers like photonic crystal fibers and antiresonant fibers [158], are promising candidates. The improvement of these system components also constitutes a fundamental foundation for the advancement of the optical sensing.

Sixth, the production cost of high-precision IFOGs remains high, and product consistency often depends on the manufacturer's technical capabilities. Performance variations may occur across different production batches. The photonic integration of components for emission, splitting, modulation, and detection, coupled with Application-Specific Integrated Circuit (ASIC) technology, allows for mass production of IFOGs. The integrated approach ensures superior consistency, lower cost, and a more compact form factor. Consequently, integration technology represents a pivotal direction for the future development of high-precision IFOGs.

These are the directions we believe hold promise for future research. With the assistance of various advanced techniques, high-precision IFOG will demonstrate superior performance. As Lefevre stated in [159], "the fiber-optic gyro is clearly seen as a unique technology for the most demanding inertial navigation applications." In the realm of inertial sensing, ultra-high-precision IFOGs continue to exhibit their unrivaled advantages in surpassing the application boundaries: deep seas, deep space and deep underground.

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Advances in High-Precision Interferometric Fiber Optic Gyroscope

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